

模型表现与 Sonnet 5 Verifier 审计

10 个 Hard case • 6 个 level • 4 个 task • 11 个被测模型 • 2,640 条 evaluation

核心判断：当前 leaderboard 只能称为“Sonnet 5 judge 下的观测排名”。L1 已被系统性 false negative 污染，L3/L4 存在任务—模态失配，L5/L6 又被 Blender 执行率与遮挡显著支配。

2,640	402	84	7+	3
verdicts	render failures	首次空/不可解析 judge	同像素相反 verdict 组	抽样确认 false positives

研究方法：每个 Level 至少一个 GPT-5.6 Sol 独立 agent；三组跨层全量交叉审计；false-negative、false-positive 与 verifier 协议专项复核；脚本、error log、六视图和重复判定交叉验证。

生成日期：2026-07-20 • America/New_York

目录

目录	2
1 • 执行摘要	4
2 • 范围、口径与研究方法	6
3 • 按 Level 审计	7
Level 1 • 单参数修改	7
Level 2 • 结构性局部修改	9
Level 3 • 多参数组合修改	11
Level 4 • 拓扑与布尔修改	13
Level 5 • 风格与语义改造	15
Level 6 • 复杂场景构建	17
4 • 按模型审计	19
Claude Opus 4.8 • observed rank #1	20
Claude Sonnet 5 • observed rank #2	21
GPT-5.5 • observed rank #3	22
Claude Sonnet 4.6 • observed rank #4	23
GPT-5.6 Terra • observed rank #5	24
Claude Haiku 4.5 • observed rank #6	25
gpt-oss-120b • observed rank #7	26
GPT-5.6 Sol • observed rank #8	27
Gemma 4 31B • observed rank #9	28
GPT-5.6 Luna • observed rank #10	29
Qwen3-Coder-Next • observed rank #11	30

5 • Sonnet 5 Verifier 专项审计	31
6 • 执行失败与模型稳定性矩阵	36
附录 A • 60 个任务的 observed 难度	37
Level 1 任务矩阵	37
Level 2 任务矩阵	40
Level 3 任务矩阵	43
Level 4 任务矩阵	46
Level 5 任务矩阵	49
Level 6 任务矩阵	52
附录 B • Agent fan-out 与证据清单	56

1 • 执行摘要

不要把 observed score 当作 ground truth。Opus 4.8 的 raw overall 最高（0.752），Sonnet 5 次之（0.727）；但各 Level 测量对象不同，且 verifier 已被证实存在 false negative、false positive 和不可判任务。

原始总分与执行成功率

#	模型	Overall	L1	L2	L3	L4	L5	L6	RF	出图率
1	Claude Opus 4.8	0.752	0.800	0.838	0.825	0.602	0.825	0.622	8	96.7%
2	Claude Sonnet 5	0.727	0.800	0.787	0.758	0.596	0.842	0.581	3	98.8%
3	GPT-5.5	0.669	0.800	0.838	0.750	0.683	0.475	0.470	36	85.0%
4	Claude Sonnet 4.6	0.642	0.775	0.787	0.742	0.606	0.658	0.282	26	89.2%
5	GPT-5.6 Terra	0.601	0.800	0.812	0.775	0.613	0.490	0.114	53	77.9%
6	Claude Haiku 4.5	0.599	0.800	0.713	0.725	0.521	0.540	0.299	29	87.9%
7	gpt-oss-120b	0.594	0.800	0.812	0.758	0.560	0.448	0.184	27	88.8%
8	GPT-5.6 Sol	0.583	0.775	0.838	0.700	0.537	0.483	0.163	56	76.7%
9	Gemma 4 31B	0.576	0.800	0.750	0.758	0.527	0.406	0.216	34	85.8%
10	GPT-5.6 Luna	0.564	0.750	0.812	0.775	0.625	0.371	0.051	59	75.4%
11	Qwen3-Coder-Next	0.472	0.800	0.775	0.683	0.362	0.125	0.085	71	70.4%

各 Level 的有效性结论

Level	Observed mean	Full	Partial	Rendered zero	RF	有效性判断
L1 • 单参数修改	0.791	348	0	92	0	严重污染：92 个 0 分全部完成了目标 literal/参数替换，且无 render failure；分数主要反映独立归一化、低可见性、名词错误和 verifier false negative。
L2 • 结构性局部修改	0.797	343	15	79	3	部分可用：能识别部分真实几何失败，但混入完全相同像素不同判分、被遮挡细节和 checklist 歧义；97 个非满分中仅 3 个 render failure。
L3 • 多参数组合修改	0.750	228	189	19	4	需混合验证：参数/命名等非视觉属性被六视图裁判系统性漏判；多个全模型正确编辑仍被 10~11/11 判 no。
L4 • 拓扑与布尔修改	0.567	221	58	141	20	视觉单裁失配：merge、贯穿孔、布尔 union 等拓扑事实无法仅凭外观可靠判断；已有执行 EXACT Boolean UNION 却 0 分的确定案例。

Level	Observed mean	Full	Partial	Rendered zero	RF	有效性判断
L5 · 风格与语义改造	0.515	187	74	13	166	需二裁：失败由 Blender/API 执行问题与主观风格/可识别性共同组成；部分 recognizability 项存在高置信 false negative。
L6 · 复杂场景构建	0.279	41	153	37	209	原始分不可直排：大量模型先被执行成功率支配；场景遮挡又造成 verifier 猜测。Gemma 唯一 full pass 是高置信 false positive。

已经能够定性的结论

- L1 的 92 个零分全部完成目标参数修改；当前 L1 基本测的是 verifier/可见性，而非模型编辑能力。
- 至少 7 组六视图 RGBA 像素完全相同，却获得不同分数；最极端为 11 模型同图、3 pass / 8 fail。
- L3 对象 rename、L4 Boolean/mesh merge 等任务无法由 image-only verifier 可靠判定。
- 在 16 个 L5/L6 满分分层抽样中，确认 3 个 false positive、1 个疑似过宽；不能外推总体误判率。
- Sol/Terra/Luna/Qwen 的高层 raw score 主要被 Blender/API 执行率支配；成功出图后的条件质量需另报。
- Sonnet 5 可用作初筛，但不应继续充当无置信度、无解释、无 abstain 的单裁 ground truth。

2 · 范围、口径与研究方法

- 主批次：benchmark/results/ReconstructedBPY_hardest10_le400_q4_sonnet5。
- 覆盖：10 jobs $\times$ 6 levels $\times$ 4 queries $\times$ 11 models = 2,640 evaluations；每模型 240 条。
- 证据：query/checklist、模型生成代码、实际执行脚本、render_results、error log、原始/修改后六视图、evaluation yes/no/score。
- 分层 agent：Level 1–6 各由一个显式 GPT-5.6 Sol agent 独立审计。
- 交叉 agent：L1–2、L3–4、L5–6 三组全量审计；另有 verifier false-negative、false-positive 和协议专项。
- 分数口径：full=score 1；partial=0<score<1；rendered zero 与 render failure 分开；rendered-only mean 排除 RF。

失败分类

类别	含义	典型证据
E · 执行/渲染	脚本不能在当前 Blender 环境完成	Attribute/Type/Key/Name/Syntax/Runtime 等 error log
G · 几何/逻辑	已出图，但数量、位置、形状、关系或 Boolean 结果不满足	代码 + 六视图可见失败
S · 语义/风格	核心物体、风格、道具、尺度或场景连贯性不足	L5/L6 rendered non-full
O · 不可观察	代码可能完成，但当前六视图被遮挡、sub-pixel 或无剖切	内部 pocket、column、封闭房间
V · Verifier 错误	相同像素不同 verdict，或图像证据清楚仍判反	像素 hash、5 $\times$ 重跑、bbox 量化
A · 任务歧义/无效	query 与源代码语义不一致，或 checklist 要求非视觉 metadata	DEPTH alias、object rename

采样参数校正：部分 agent 底稿把配置默认 temperature=.6 表述成实际请求值；当前 adaptive Sonnet 5 transport 会省略 temperature。运行产物没有固化有效请求参数、seed 与代码 commit，因此实际非确定性来源不可复现。

3 · 按 Level 审计

每个 Level 均给出全量统计、11 模型代表成功/失败、任务难度两端和独立审计结论。

Level 1 · 单参数修改

严重污染 · 92 个 0 分全部完成了目标 literal/参数替换，且无 render failure；分数主要反映独立归一化、低可见性、名词错误和 verifier false negative。

0.791
observed mean

348
full / 440

0
partial

92
rendered zero

0
render failure

独立审计结论

- 92 个 observed zero 全部完成目标 literal/参数替换，且没有 render failure。
- 59 个是明确 false negative，11 个来自无效任务语义，22 个低于可靠视觉可见尺度或被遮挡。
- sofa 高度、sofa 长度、stair width 三项用当前 Sonnet 5 各复测 5 次，仍全部判 0；蓝色控制项 5/5 判 1。
- 结论：修复任务定义和 verifier 前，不应使用 Level 1 排名。

模型统计与代表案例

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Claude Opus 4.8	0.800	32/0/8/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
Claude Sonnet 4.6	0.775	31/0/9/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
Claude Sonnet 5	0.800	32/0/8/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
Claude Haiku 4.5	0.800	32/0/8/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
GPT-5.6 Sol	0.775	31/0/9/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
GPT-5.6 Terra	0.800	32/0/8/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
GPT-5.6 Luna	0.750	30/0/10/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
GPT-5.5	0.800	32/0/8/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—
gpt-oss-120b	0.800	32/0/8/0	0005 · L1Q0 · ceiling-mounting-bracket · 1.000	0060 · L1Q2 · mini-storage-drawers · 0.000 · 风险：任务语义无效 + 独立归一化	—

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Gemma 4 31B	0.800	32/0/8/0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险: 任务语义无效 + 独立归一化	—
Qwen3-Coder-Next	0.800	32/0/8/0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险: 任务语义无效 + 独立归一化	—

最低 / 最高 observed mean 的任务

端	Case	Query	Mean	F/P/Z/RF	审计风险
低	0060 • L1Q2 • mini-storage-drawers	Increase the depth (DEPTH) of the cabinet from 68.0 to 90.0.	0.000	0/0/11/0	任务语义无效 + 独立归一化
低	0060 • L1Q3 • mini-storage-drawers	Change the bottom rail thickness (BOTTOM_RAIL_THICKNESS) from 1.0 to 3.0.	0.000	0/0/11/0	目标被外壳遮挡
低	0061 • L1Q1 • customizable-furniture-minis	Increase the armchair's overall height from 34 to 50.	0.000	0/0/11/0	高置信 false negative
低	0061 • L1Q2 • customizable-furniture-minis	Increase the sofa length from 54 to 80.	0.000	0/0/11/0	高置信 false negative
低	0095 • L1Q3 • helical-staircase	Increase the step_width parameter from 24.0 to 35.0 to make the treads extend further outward.	0.000	0/0/11/0	高置信 false negative
低	0101 • L1Q2 • extrusion-2020	Increase the core bore diameter from 4.2mm to 6mm.	0.000	0/0/11/0	高置信 false negative
高	0005 • L1Q0 • ceiling-mounting-bracket	Change the bracket color from yellow to blue.	1.000	11/0/0/0	—
高	0005 • L1Q1 • ceiling-mounting-bracket	Increase the mounting screw radius from 2.5 to 4.0.	1.000	11/0/0/0	—
高	0005 • L1Q2 • ceiling-mounting-bracket	Reduce the wall thickness from 2.0 to 1.0.	1.000	11/0/0/0	—
高	0011 • L1Q0 • curved-corner-bracket	Change the bracket color from yellow to blue.	1.000	11/0/0/0	—

Level 2 · 结构性局部修改

部分可用 · 能识别部分真实几何失败，但混入完全相同像素不同判分、被遮挡细节和 checklist 歧义；97 个非满分中仅 3 个 render failure。

0.797
observed mean

343
full / 440

15
partial

79
rendered zero

3
render failure

独立审计结论

- 440 条中 343 full、11 partial、83 rendered zero、3 render failure。
- Level 2 能识别部分真实结构错误，但混入 primitive/截面/细孔不可见和同像素 verdict 翻转。
- spiral-stairs L2Q3 的 11 组六视图完全相同，却分裂为 3 pass / 8 fail。
- 需要 close-up/section view，并将 primitive 类型、孔数、循环角度交给 scene/geometry verifier。

模型统计与代表案例

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Claude Opus 4.8	0.838	33/1/6/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
Claude Sonnet 4.6	0.787	31/1/7/1	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	0005 · L2Q2 · ceiling-mounting-bracket · 0.000
Claude Sonnet 5	0.787	31/1/8/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
Claude Haiku 4.5	0.713	27/3/10/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
GPT-5.6 Sol	0.838	33/1/6/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
GPT-5.6 Terra	0.812	32/1/7/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
GPT-5.6 Luna	0.812	32/1/7/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
GPT-5.5	0.838	33/1/6/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
gpt-oss-120b	0.812	32/1/6/1	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	0101 · L2Q3 · extrusion-2020 · 0.000 · 风险：细节不可可靠视觉辨认
Gemma 4 31B	0.750	29/2/9/0	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	—
Qwen3-Coder-Next	0.775	30/2/7/1	0005 · L2Q0 · ceiling-mounting-bracket · 1.000	0011 · L2Q3 · curved-corner-bracket · 0.000 · 风险：内容被几何遮挡	0005 · L2Q2 · ceiling-mounting-bracket · 0.000

最低 / 最高 observed mean 的任务

端	Case	Query	Mean	F/P/Z/RF	审计风险
低	0101 • L2Q3 • extrusion-2020	Replace the four thin rectangular diagonal relief channels with small circular holes instead.	0.000	0/0/10/1	细节不可靠视觉辨认
低	0011 • L2Q3 • curved-corner-bracket	Remove the corner fill block so the two straight legs meet with an open gap at the inner corner instead of being j...	0.000	0/0/11/0	内容被几何遮挡
低	0028 • L2Q0 • parametric-spiral-stairs	Replace the central column with a rectangular box pillar instead of a cylinder.	0.000	0/0/11/0	中心柱被踏步遮挡
低	0028 • L2Q2 • parametric-spiral-stairs	Change the tread shape so each step is a full rectangular slab spanning the entire radius width instead of a pie-s...	0.091	1/0/10/0	—
低	0028 • L2Q3 • parametric-spiral-stairs	Make the staircase spiral in the opposite rotational direction (reverse the winding direction).	0.273	3/0/8/0	11 模型同像素、3/8 分裂
低	0005 • L2Q3 • ceiling-mounting-bracket	Reorient the mounting screw hole and boss so the hole axis points vertically (up/down through the top) instead of ...	0.364	4/0/7/0	—
高	0005 • L2Q0 • ceiling-mounting-bracket	Add a second mounting screw hole (with its boss) next to the existing one, positioned side by side along the width...	1.000	11/0/0/0	—
高	0011 • L2Q0 • curved-corner-bracket	Increase the number of small mounting holes along each straight leg from 3 to 5.	1.000	11/0/0/0	—
高	0011 • L2Q2 • curved-corner-bracket	Make the circular segment resolution much lower so the curved hoop looks faceted/polygonal instead of smooth.	1.000	11/0/0/0	—
高	0028 • L2Q1 • parametric-spiral-stairs	Reduce the number of steps per full turn from 13 to 8, making each tread cover a wider angle.	1.000	11/0/0/0	—

Level 3 · 多参数组合修改

需混合验证 · 参数/命名等非视觉属性被六视图裁判系统性漏判；多个全模型正确编辑仍被 10–11/11 判 no。

0.750	228	189	19	4
observed mean	full / 440	partial	rendered zero	render failure

独立审计结论

- 对象 rename 任务无法由 RGB 像素验证；11 个模型都完成 rename，第三 checklist 却 11/11 判 no。
- extrusion length/core diameter、spiral column/step height 等参数 11/11 改对，仍被系统性判 no。
- 独立 center/scale normalization 会削弱长度、高度和宽度变化。
- Level 3 应采用参数、metadata、scene invariant 与图像组合评分。

模型统计与代表案例

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Claude Opus 4.8	0.825	23/17/0/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.333 · 风险：对象重命名不可由像素验证	—
Claude Sonnet 4.6	0.742	22/14/2/2	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	0028 · L3Q2 · parametric-spiral-stairs · 0.000
Claude Sonnet 5	0.758	22/17/1/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.333 · 风险：对象重命名不可由像素验证	—
Claude Haiku 4.5	0.725	20/16/4/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
GPT-5.6 Sol	0.700	17/21/2/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
GPT-5.6 Terra	0.775	23/16/1/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
GPT-5.6 Luna	0.775	23/15/2/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
GPT-5.5	0.750	22/16/2/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
gpt-oss-120b	0.758	20/19/1/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
Gemma 4 31B	0.758	20/19/1/0	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	—
Qwen3-Coder-Next	0.683	16/19/3/2	0005 · L3Q1 · ceiling-mounting-bracket · 1.000	0101 · L3Q3 · extrusion-2020 · 0.000 · 风险：对象重命名不可由像素验证	0060 · L3Q1 · mini-storage-drawers · 0.000

最低 / 最高 observed mean 的任务

端	Case	Query	Mean	F/P/Z/RF	审计风险
低	0101 • L3Q3 • extrusion-2020	Remove the diagonal relief cutters entirely, shrink the internal pocket holes to a smaller diameter, and rename th...	0.061	0/2/9/0	对象重命名不可由像素验证
低	0110 • L3Q3 • cam-profile	Apply these three edits: shift the fall_start and fall_end angles so the dwell (flat) region after the lift is sho...	0.182	0/5/6/0	—
低	0095 • L3Q3 • helical-staircase	Perform three separate modifications: double the overall number of steps (step_count) to make the staircase taller...	0.333	0/11/0/0	—
低	0028 • L3Q1 • parametric-spiral-stairs	Make the central column thicker, increase the step height so the stairs are taller and more spaced out, and change...	0.364	0/11/0/0	全模型正确参数、系统性漏判
低	0101 • L3Q0 • extrusion-2020	Make three independent changes: increase the extrusion length to 100mm, enlarge the center bore diameter to 6mm, a...	0.364	0/11/0/0	全模型正确参数、系统性漏判
低	0005 • L3Q2 • ceiling-mounting-bracket	Update the model with these independent changes: make the rear screw boss faceted (low-poly, like an octagon) inst...	0.485	0/11/0/0	—
高	0005 • L3Q1 • ceiling-mounting-bracket	Modify the bracket in three ways: make the overall bracket body taller, reduce the standoff height so there is les...	1.000	11/0/0/0	—
高	0011 • L3Q0 • curved-corner-bracket	Make three independent changes to the bracket: increase the number of small mounting holes from 3 to 5, completely...	1.000	11/0/0/0	—
高	0011 • L3Q1 • curved-corner-bracket	Update the bracket with these independent edits: increase the width (depth along the extrusion axis) so the bracke...	1.000	11/0/0/0	—
高	0028 • L3Q0 • parametric-spiral-stairs	Increase the staircase radius so the treads are wider, reduce the number of revolutions to 2, and change the centr...	1.000	11/0/0/0	—

Level 4 · 拓扑与布尔修改

视觉单裁失配 · merge、贯穿孔、布尔 union 等拓扑事实无法仅凭外观可靠判断；已有执行 EXACT Boolean UNION 却 0 分的确定案例。

0.567
observed mean

221
full / 440

58
partial

141
rendered zero

20
render failure

独立审计结论

- merge、mesh connectivity、through-hole、Boolean union 是拓扑事实，普通六视图只能看到外观。
- Terra 对 base 与 bottom tread 执行 EXACT Boolean UNION 并删除 operand，Sonnet 5 原判和 5 次复测均为 0。
- 反向也存在：drawers L4Q3 的多个 1.0 脚本仍返回两个独立 objects，只是视觉接触。
- Level 4 必须增加 mesh connectivity、boolean、watertight/through-hole 等确定性检查。

模型统计与代表案例

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Claude Opus 4.8	0.602	21/6/13/0	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	—
Claude Sonnet 4.6	0.606	21/6/12/1	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	0061 · L4Q0 · customizable-furniture-minis · 0.000
Claude Sonnet 5	0.596	22/4/14/0	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	—
Claude Haiku 4.5	0.521	17/8/14/1	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	0110 · L4Q0 · cam-profile · 0.000
GPT-5.6 Sol	0.537	20/3/13/4	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	0028 · L4Q3 · parametric-spiral-stairs · 0.000
GPT-5.6 Terra	0.613	22/6/12/0	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	—
GPT-5.6 Luna	0.625	23/4/13/0	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	—
GPT-5.5	0.683	25/5/9/1	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	0061 · L4Q0 · customizable-furniture-minis · 0.000
gpt-oss-120b	0.560	20/5/15/0	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	—
Gemma 4 31B	0.527	18/6/13/3	0129 · L4Q0 · neo-strip · 1.000	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证	0005 · L4Q2 · ceiling-mounting-bracket · 0.000
Qwen3-Coder-Next	0.362	12/5/13/10	0129 · L4Q0 · neo-strip · 1.000	0095 · L4Q2 · helical-staircase · 0.000 · 风险：Boolean union 拓扑不可由外观验证	0011 · L4Q2 · curved-corner-bracket · 0.000 · 风险：merge 拓扑不可由外观验证

最低 / 最高 observed mean 的任务

端	Case	Query	Mean	F/P/Z/RF	审计风险
低	0011 • L4Q2 • curved-corner-bracket	Merge the two straight legs and the corner fill block into a single continuous L-shaped piece by rounding the inner...	0.000	0/0/10/1	merge 拓扑不可由外观验证
低	0028 • L4Q1 • parametric-spiral-stairs	Cut a circular hole through the center of every tread so the central column passes cleanly through each step witho...	0.000	0/0/10/1	—
低	0095 • L4Q2 • helical-staircase	Merge the base disk and the bottom-most tread into a single continuous solid piece so there is no visible seam bet...	0.000	0/0/11/0	Boolean union 拓扑不可由外观验证
低	0028 • L4Q3 • parametric-spiral-stairs	Merge the central column and all the treads into a single continuous solid piece.	0.091	1/0/8/2	—
低	0110 • L4Q0 • cam-profile	Cut a radial set-screw hole through the wall of the raised hub boss that breaks through into the central shaft bo...	0.091	1/0/8/2	—
低	0064 • L4Q3 • tool-shelf-for-bench	Add a raised front lip along the top front edge of the front-level shelf plate to keep small tools from rolling of...	0.091	1/0/10/0	—
高	0129 • L4Q0 • neo-strip	Add hexagonal void cutouts through the body of the strip channel at each pixel location, matching the hexagonal ho...	1.000	11/0/0/0	—
高	0110 • L4Q3 • cam-profile	Cut a small flat notch into the outer rim of the cam profile at the point of maximum lift to act as a timing index...	0.939	10/1/0/0	—
高	0064 • L4Q1 • tool-shelf-for-bench	Merge the three separate center support walls (rear, mid, front) into a single continuous center divider panel tha...	0.909	9/2/0/0	—
高	0129 • L4Q1 • neo-strip	Cut mounting screw holes through the base of the strip channel body, positioned between the divider slots, so the ...	0.909	10/0/1/0	—

Level 5 · 风格与语义改造

需二裁 · 失败由 Blender/API 执行问题与主观风格/可识别性共同组成；部分 recognizability 项存在高置信 false negative。

0.515
observed mean

187
full / 440

74
partial

13
rendered zero

166
render failure

独立审计结论

- observed mean 同时混合 Blender/API 执行失败和主观风格/可识别性评分。
- Qwen aged staircase 仅使用干净纯色金属，却在 aged/weathered 项被判 yes，是确定 false pass。
- Gemma 与 gpt-oss organic bracket 保留清楚的夹持通道和 boss/hole，却在 recognizability 项被判 no。
- 风格项需要视觉 anchor、复合 checklist conjunction 和跨家族二裁。

模型统计与代表案例

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Claude Opus 4.8	0.825	29/6/0/5	0011 · L5Q2 · curved-corner-bracket · 1.000	0095 · L5Q1 · helical-staircase · 0.667	0005 · L5Q1 · ceiling-mounting-bracket · 0.000
Claude Sonnet 4.6	0.658	22/7/1/10	0011 · L5Q2 · curved-corner-bracket · 1.000	0095 · L5Q1 · helical-staircase · 0.000	0005 · L5Q2 · ceiling-mounting-bracket · 0.000
Claude Sonnet 5	0.842	28/9/2/1	0005 · L5Q0 · ceiling-mounting-bracket · 1.000	0095 · L5Q0 · helical-staircase · 0.000	0028 · L5Q3 · parametric-spiral-stairs · 0.000
Claude Haiku 4.5	0.540	16/11/1/12	0011 · L5Q2 · curved-corner-bracket · 1.000	0095 · L5Q1 · helical-staircase · 0.000	0005 · L5Q2 · ceiling-mounting-bracket · 0.000
GPT-5.6 Sol	0.483	18/2/0/20	0011 · L5Q2 · curved-corner-bracket · 1.000	0101 · L5Q2 · extrusion-2020 · 0.667	0011 · L5Q0 · curved-corner-bracket · 0.000
GPT-5.6 Terra	0.490	18/3/0/19	0011 · L5Q2 · curved-corner-bracket · 1.000	0101 · L5Q2 · extrusion-2020 · 0.333	0028 · L5Q0 · parametric-spiral-stairs · 0.000 · 风险：风格阈值存在 false-pass 风险
GPT-5.6 Luna	0.371	14/2/2/22	0011 · L5Q2 · curved-corner-bracket · 1.000	0095 · L5Q1 · helical-staircase · 0.000	0005 · L5Q1 · ceiling-mounting-bracket · 0.000
GPT-5.5	0.475	19/0/2/19	0011 · L5Q2 · curved-corner-bracket · 1.000	0028 · L5Q1 · parametric-spiral-stairs · 0.000	0005 · L5Q0 · ceiling-mounting-bracket · 0.000
gpt-oss-120b	0.448	10/16/2/12	0005 · L5Q0 · ceiling-mounting-bracket · 1.000	0028 · L5Q0 · parametric-spiral-stairs · 0.250 · 风险：风格阈值存在 false-pass 风险	0011 · L5Q0 · curved-corner-bracket · 0.000

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Gemma 4 31B	0.406	10/12/2/16	0011 • L5Q2 • curved-corner-bracket • 1.000	0028 • L5Q0 • parametric-spiral-stairs • 0.750 • 风险：风格阈值存在 false-pass 风险	0005 • L5Q1 • ceiling-mounting-bracket • 0.000
Qwen3-Coder-Next	0.125	3/6/1/30	0110 • L5Q1 • cam-profile • 1.000	0061 • L5Q2 • customizable-furniture-minis • 0.000	0005 • L5Q0 • ceiling-mounting-bracket • 0.000

最低 / 最高 observed mean 的任务

端	Case	Query	Mean	F/P/Z/RF	审计风险
低	0095 • L5Q1 • helical-staircase	Transform the staircase into a medieval castle tower stairway, giving it a heavy stone-fortress aesthetic.	0.152	0/3/4/4	—
低	0095 • L5Q0 • helical-staircase	Reimagine this staircase as an ancient DNA double-helix sculpture, evoking genetic strands intertwining upward.	0.212	0/4/1/6	—
低	0101 • L5Q1 • extrusion-2020	Reinterpret the extrusion with a futuristic, sci-fi spacecraft-panel aesthetic.	0.273	3/0/0/8	—
低	0028 • L5Q3 • parametric-spiral-stairs	Make the spiral staircase feel like it belongs in a haunted, decaying gothic mansion.	0.333	3/2/0/6	—
低	0064 • L5Q1 • tool-shelf-for-bench	Give this tool shelf an organic, nature-inspired look, as if it grew from a tree trunk with wood-grain surfaces and...	0.333	2/3/0/6	—
低	0110 • L5Q2 • cam-profile	Transform the cam profile into an organic, biomechanical form, as if it were grown rather than machined.	0.333	2/4/1/4	—
高	0011 • L5Q2 • curved-corner-bracket	Redesign the bracket to feel lightweight and aerodynamic, like a component from a race car or aircraft.	0.795	8/2/0/1	—
高	0129 • L5Q2 • neo-strip	Give this socket strip a stealthy, military-grade tactical look.	0.788	8/2/0/1	—
高	0005 • L5Q0 • ceiling-mounting-bracket	Redesign the bracket with an industrial heavy-duty aesthetic, as if it belongs in a factory or warehouse.	0.788	8/1/0/2	—
高	0064 • L5Q2 • tool-shelf-for-bench	Transform this tool shelf into a sleek, minimalist Scandinavian-style storage piece.	0.788	8/1/0/2	—

Level 6 · 复杂场景构建

原始分不可直排 · 大量模型先被执行成功率支配；场景遮挡又造成 verifier 猜测。Gemma 唯一 full pass 是高置信 false positive。

0.279
observed mean

41
full / 440

153
partial

37
rendered zero

209
render failure

独立审计结论

- Sol/Terra/Luna 仅 8/6/3 条成功出图；raw mean 主要被 Blender API 兼容性支配。
- 封闭外壳与 room shell 常遮挡内部场景，verifier 会在不可见内容上猜测。
- Gemma lighthouse 与 Haiku electronics workbench 均获 7/7，独立复核分别只支持 4/7 与 2/7。
- 应增加 interior/cutaway camera，并分开发布出图率、rendered-only mean 和语义完成度。

模型统计与代表案例

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
Claude Opus 4.8	0.622	6/29/2/3	0129 · L6Q0 · neo-strip · 1.000 · 风险：对象存在不等于关系/尺度成立	0028 · L6Q0 · parametric-spiral-stairs · 0.571 · 风险：封闭圆筒遮挡内部；存在 false pass	0011 · L6Q2 · curved-corner-bracket · 0.000
Claude Sonnet 4.6	0.282	4/15/9/12	0129 · L6Q0 · neo-strip · 1.000 · 风险：对象存在不等于关系/尺度成立	0028 · L6Q0 · parametric-spiral-stairs · 0.429 · 风险：封闭圆筒遮挡内部；存在 false pass	0011 · L6Q3 · curved-corner-bracket · 0.000
Claude Sonnet 5	0.581	7/29/2/2	0064 · L6Q0 · tool-shelf-for-bench · 1.000	0028 · L6Q0 · parametric-spiral-stairs · 0.429 · 风险：封闭圆筒遮挡内部；存在 false pass	0095 · L6Q1 · helical-staircase · 0.000
Claude Haiku 4.5	0.299	1/19/4/16	0129 · L6Q0 · neo-strip · 1.000 · 风险：对象存在不等于关系/尺度成立	0028 · L6Q0 · parametric-spiral-stairs · 0.000 · 风险：封闭圆筒遮挡内部；存在 false pass	0005 · L6Q1 · ceiling-mounting-bracket · 0.000
GPT-5.6 Sol	0.163	5/3/0/32	0129 · L6Q0 · neo-strip · 1.000 · 风险：对象存在不等于关系/尺度成立	0028 · L6Q0 · parametric-spiral-stairs · 0.429 · 风险：封闭圆筒遮挡内部；存在 false pass	0005 · L6Q0 · ceiling-mounting-bracket · 0.000
GPT-5.6 Terra	0.114	4/1/1/34	0064 · L6Q0 · tool-shelf-for-bench · 1.000	0095 · L6Q3 · helical-staircase · 0.000	0005 · L6Q0 · ceiling-mounting-bracket · 0.000
GPT-5.6 Luna	0.051	0/3/0/37	—	0095 · L6Q0 · helical-staircase · 0.500	0005 · L6Q0 · ceiling-mounting-bracket · 0.000
GPT-5.5	0.470	13/10/1/16	0129 · L6Q0 · neo-strip · 1.000 · 风险：对象存在不等于关系/尺度成立	0095 · L6Q3 · helical-staircase · 0.000	0005 · L6Q2 · ceiling-mounting-bracket · 0.000

模型	Mean	F/P/Z/RF	代表成功	代表 observed 非满分	代表执行失败
gpt-oss-120b	0.184	0/19/7/14	—	0028 • L6Q0 • parametric-spiral-stairs • 0.429 • 风险：封闭圆筒遮挡内部；存在 false pass	0005 • L6Q1 • ceiling-mounting-bracket • 0.000
Gemma 4 31B	0.216	1/16/8/15	0028 • L6Q0 • parametric-spiral-stairs • 1.000 • 风险：封闭圆筒遮挡内部；存在 false pass	0129 • L6Q3 • neo-strip • 0.000	0005 • L6Q2 • ceiling-mounting-bracket • 0.000
Qwen3-Coder-Next	0.085	0/9/3/28	—	0129 • L6Q0 • neo-strip • 0.429 • 风险：对象存在不等于关系/尺度成立	0005 • L6Q1 • ceiling-mounting-bracket • 0.000

最低 / 最高 observed mean 的任务

端	Case	Query	Mean	F/P/Z/RF	审计风险
低	0095 • L6Q1 • helical-staircase	Set the helical staircase in an outdoor garden courtyard: surround the base with a circular stone patio, add potte...	0.015	0/1/0/10	—
低	0095 • L6Q3 • helical-staircase	Create a fantasy tower library scene with the staircase as the centerpiece: add curved bookshelves lining the surr...	0.015	0/1/6/4	—
低	0011 • L6Q2 • curved-corner-bracket	Build a furniture assembly scene: use the curved corner bracket to reinforce the corner joint between a wooden tab...	0.091	0/3/2/6	—
低	0110 • L6Q2 • cam-profile	Set the cam profile inside an engine valve-timing assembly scene: place it on a rotating camshaft supported by two...	0.152	0/4/1/6	—
低	0129 • L6Q3 • neo-strip	Place the strip channel in an outdoor patio decoration scene: mount it along the underside of a wooden pergola bea...	0.156	0/2/2/7	—
低	0011 • L6Q3 • curved-corner-bracket	Design an outdoor pergola support scene: mount the curved corner bracket at the junction of a vertical wooden post...	0.169	1/1/0/9	—
高	0129 • L6Q0 • neo-strip	Set up a workbench scene around this NeoPixel strip channel: place it on a wooden workbench surface, add a spool o...	0.610	5/3/0/3	对象存在不等于关系/尺度成立
高	0064 • L6Q0 • tool-shelf-for-bench	Mount the tool shelf onto the back wall above a wooden workbench: add a rectangular workbench surface positioned d...	0.532	4/4/1/2	—
高	0060 • L6Q0 • mini-storage-drawers	Place the mini storage drawer cabinet on top of a wooden desk in a home office. Add a desk lamp beside it casting ...	0.481	4/3/0/4	—
高	0095 • L6Q0 • helical-staircase	Place the helical staircase inside a modern multi-story atrium: add a floor slab beneath it, a ceiling skylight ab...	0.470	3/5/0/3	—

4 • 按模型审计

以下每个模型均拆分 Level score、full/partial/zero/RF、异常类型与代表案例。风险标签表示 observed 非满分可能来自 verifier 或可见性，不应直接归因模型。

Claude Opus 4.8 • observed rank #1

0.752 raw overall	232 成功出图	8 render failures	96.7% 出图率
----------------------	-------------	----------------------	--------------

失败类别

- L1-L3：正确参数/细节被视觉裁判漏判
 - L4：复杂拓扑与多条件部分完成
 - L5：少量 Blender/API 执行失败与风格可识别性
 - L6：道具 completeness、遮挡、尺度与场景连贯性
- RF exception class: AttributeError 2; 其他/无日志 2; TypeError 2; SyntaxError 1; ValueError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.838	0.838	33	1	6	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.825	0.825	23	17	0	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.333 • 风险：对象重命名不可由像素验证
L4	0.602	0.602	21	6	13	0	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证
L5	0.825	0.943	29	6	0	5	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0095 • L5Q1 • helical-staircase • 0.667 RF: 0005 • L5Q1 • ceiling-mounting-bracket • 0.000
L6	0.622	0.673	6	29	2	3	0129 • L6Q0 • neo-strip • 1.000 • 风险：对象存在不等于关系/尺度成立 Set up a workbench scene around this NeoPixel strip channel: place it on a wooden...	0028 • L6Q0 • parametric-spiral-stairs • 0.571 • 风险：封闭圆筒遮挡内部；存在 false pass RF: 0011 • L6Q2 • curved-corner-bracket • 0.000

Claude Sonnet 5 • observed rank #2

0.727 raw overall	237 成功出图	3 render failures	98.8% 出图率
----------------------	-------------	----------------------	--------------

失败类别

- 低层：与其他模型相同的 verifier false negative
 - L3/L4：非视觉属性、拓扑事实与 checklist 歧义
 - L5：极少执行失败，主要是风格/几何语义未完成
 - L6：复杂场景 completeness 和内容可见性
- RF exception class: AttributeError 2; TypeError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.787	0.787	31	1	8	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.758	0.758	22	17	1	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.333 • 风险：对象重命名不可由像素验证
L4	0.596	0.596	22	4	14	0	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证
L5	0.842	0.863	28	9	2	1	0005 • L5Q0 • ceiling-mounting-bracket • 1.000 Redesign the bracket with an industrial heavy-duty aesthetic, as if it belongs in...	0095 • L5Q0 • helical-staircase • 0.000 RF: 0028 • L5Q3 • parametric-spiral-stairs • 0.000
L6	0.581	0.611	7	29	2	2	0064 • L6Q0 • tool-shelf-for-bench • 1.000 Mount the tool shelf onto the back wall above a wooden workbench: add a rectangul...	0028 • L6Q0 • parametric-spiral-stairs • 0.429 • 风险：封闭圆筒遮挡内部；存在 false pass RF: 0095 • L6Q1 • helical-staircase • 0.000

GPT-5.5 • observed rank #3

0.669 raw overall	204 成功出图	36 render failures	85.0% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- 低层：裁判误判/低可见性为主
 - L4：少量真实几何或执行失败
 - L5/L6：明显双峰——API 执行失败多，跑成后条件均分高
 - 复杂场景：遮挡、道具 completeness 与视觉证据不足
- RF exception class: AttributeError 32; TypeError 4

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.838	0.838	33	1	6	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.750	0.750	22	16	2	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.683	0.701	25	5	9	1	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证 RF: 0061 • L4Q0 • customizable-furniture-minis • 0.000
L5	0.475	0.905	19	0	2	19	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0028 • L5Q1 • parametric-spiral-stairs • 0.000 RF: 0005 • L5Q0 • ceiling-mounting-bracket • 0.000
L6	0.470	0.783	13	10	1	16	0129 • L6Q0 • neo-strip • 1.000 • 风险：对象存在不等于关系/尺度成立 Set up a workbench scene around this NeoPixel strip channel: place it on a wooden...	0095 • L6Q3 • helical-staircase • 0.000 RF: 0005 • L6Q2 • ceiling-mounting-bracket • 0.000

Claude Sonnet 4.6 • observed rank #4

0.642 raw overall	214 成功出图	26 render failures	89.2% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- 低层：可见性/裁判误判与少量真实几何缺陷
 - 中高层：Syntax/Type/Key/Attribute 等执行失败
 - L5：风格语义不完整或核心物体失去可识别性
 - L6：复杂场景缺件、尺度、遮挡与连贯性
- RF exception class: KeyError 10; TypeError 9; AttributeError 5; SyntaxError 1; RuntimeError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.775	0.775	31	0	9	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.787	0.808	31	1	7	1	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡 RF: 0005 • L2Q2 • ceiling-mounting-bracket • 0.000
L3	0.742	0.781	22	14	2	2	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证 RF: 0028 • L3Q2 • parametric-spiral-stairs • 0.000
L4	0.606	0.622	21	6	12	1	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证 RF: 0061 • L4Q0 • customizable-furniture-minis • 0.000
L5	0.658	0.878	22	7	1	10	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0095 • L5Q1 • helical-staircase • 0.000 RF: 0005 • L5Q2 • ceiling-mounting-bracket • 0.000
L6	0.282	0.403	4	15	9	12	0129 • L6Q0 • neo-strip • 1.000 • 风险：对象存在不等于关系/尺度成立 Set up a workbench scene around this NeoPixel strip channel: place it on a wooden...	0028 • L6Q0 • parametric-spiral-stairs • 0.429 • 风险：封闭圆筒遮挡内部；存在 false pass RF: 0011 • L6Q3 • curved-corner-bracket • 0.000

GPT-5.6 Terra • observed rank #5

0.601 raw overall	187 成功出图	53 render failures	77.9% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- 低层：视觉不可验证及相同像素判分翻转
 - L4：拓扑事实被漏判，也有复杂编辑不完整
 - L5/L6 主因：Attribute/Type/API 兼容性失败
 - 成功渲染后：有机化、场景配件和可见性不足
- RF exception class: AttributeError 28; TypeError 25

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.812	0.812	32	1	7	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.775	0.775	23	16	1	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.613	0.613	22	6	12	0	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证
L5	0.490	0.933	18	3	0	19	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0101 • L5Q2 • extrusion-2020 • 0.333 RF: 0028 • L5Q0 • parametric-spiral-stairs • 0.000 • 风险：风格阈值存在 false-pass 风险
L6	0.114	0.762	4	1	1	34	0064 • L6Q0 • tool-shelf-for-bench • 1.000 Mount the tool shelf onto the back wall above a wooden workbench: add a rectangul...	0095 • L6Q3 • helical-staircase • 0.000 RF: 0005 • L6Q0 • ceiling-mounting-bracket • 0.000

Claude Haiku 4.5 • observed rank #6

0.599 raw overall	211 成功出图	29 render failures	87.9% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- L2-L4：局部几何、数量、位置或拓扑只完成一部分
 - 高层：Blender API/类型/材质 socket 执行失败
 - L5：主题/材质风格不完整或失去可识别性
 - L6：道具缺失、遮挡、尺度与场景连贯性
- RF exception class: AttributeError 12; TypeError 9; KeyError 4; RuntimeError 3; IndentationError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.713	0.713	27	3	10	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.725	0.725	20	16	4	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.521	0.534	17	8	14	1	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证 RF: 0110 • L4Q0 • cam-profile • 0.000
L5	0.540	0.771	16	11	1	12	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0095 • L5Q1 • helical-staircase • 0.000 RF: 0005 • L5Q2 • ceiling-mounting-bracket • 0.000
L6	0.299	0.498	1	19	4	16	0129 • L6Q0 • neo-strip • 1.000 • 风险：对象存在不等于关系/尺度成立 Set up a workbench scene around this NeoPixel strip channel: place it on a wooden...	0028 • L6Q0 • parametric-spiral-stairs • 0.000 • 风险：封闭圆筒遮挡内部；存在 false pass RF: 0005 • L6Q1 • ceiling-mounting-bracket • 0.000

gpt-oss-120b • observed rank #7

0.594 raw overall	213 成功出图	27 render failures	88.8% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- 低层：裁判误判、细节不可见及少量 SyntaxError
 - 中层：局部几何/拓扑部分完成
 - 高层：材质 socket、Key/Type/Attribute 与运行错误
 - L5/L6：风格语义、场景道具、尺度和连贯性
- RF exception class: KeyError 11; AttributeError 6; TypeError 4; NameError 2; RuntimeError 2; ReferenceError 1; SyntaxError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.812	0.833	32	1	6	1	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡 RF: 0101 • L2Q3 • extrusion-2020 • 0.000 • 风险：细节不可可靠视觉辨认
L3	0.758	0.758	20	19	1	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.560	0.560	20	5	15	0	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证
L5	0.448	0.640	10	16	2	12	0005 • L5Q0 • ceiling-mounting-bracket • 1.000 Redesign the bracket with an industrial heavy-duty aesthetic, as if it belongs in...	0028 • L5Q0 • parametric-spiral-stairs • 0.250 • 风险：风格阈值存在 false-pass 风险 RF: 0011 • L5Q0 • curved-corner-bracket • 0.000
L6	0.184	0.283	0	19	7	14	—	0028 • L6Q0 • parametric-spiral-stairs • 0.429 • 风险：封闭圆筒遮挡内部；存在 false pass RF: 0005 • L6Q1 • ceiling-mounting-bracket • 0.000

GPT-5.6 Sol • observed rank #8

0.583 raw overall	184 成功出图	56 render failures	76.7% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- L1-L3：多数非满分来自视觉不可验证或 verifier 误判
 - L4：复杂结构/拓扑部分完成
 - L5/L6 主因：Blender API 兼容性导致大批 render failure
 - 成功渲染后：风格细化、道具数量与整体语义偶有不足
- RF exception class: AttributeError 34; TypeError 21; KeyError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.775	0.775	31	0	9	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.838	0.838	33	1	6	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.700	0.700	17	21	2	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.537	0.597	20	3	13	4	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证 RF: 0028 • L4Q3 • parametric-spiral-stairs • 0.000
L5	0.483	0.967	18	2	0	20	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0101 • L5Q2 • extrusion-2020 • 0.667 RF: 0011 • L5Q0 • curved-corner-bracket • 0.000
L6	0.163	0.815	5	3	0	32	0129 • L6Q0 • neo-strip • 1.000 • 风险：对象存在不等于关系/尺度成立 Set up a workbench scene around this NeoPixel strip channel: place it on a wooden...	0028 • L6Q0 • parametric-spiral-stairs • 0.429 • 风险：封闭圆筒遮挡内部；存在 false pass RF: 0005 • L6Q0 • ceiling-mounting-bracket • 0.000

Gemma 4 31B • observed rank #9

0.576 raw overall	206 成功出图	34 render failures	85.8% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- 低层：局部结构真实缺陷与 verifier 翻转并存
 - 中层：几何/拓扑部分完成及执行错误
 - 高层：Attribute/Type/API 失败和主题语义不足
 - L6：场景遮挡；另有唯一 full pass 为高置信 false positive
- RF exception class: TypeError 14; AttributeError 13; KeyError 3; 其他/无日志 1; ValueError 1; ImportError 1; RuntimeError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.750	0.750	29	2	9	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.758	0.758	20	19	1	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.527	0.570	18	6	13	3	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证 RF: 0005 • L4Q2 • ceiling-mounting-bracket • 0.000
L5	0.406	0.677	10	12	2	16	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0028 • L5Q0 • parametric-spiral-stairs • 0.750 • 风险：风格阈值存在 false-pass 风险 RF: 0005 • L5Q1 • ceiling-mounting-bracket • 0.000
L6	0.216	0.345	1	16	8	15	0028 • L6Q0 • parametric-spiral-stairs • 1.000 • 风险：封闭圆筒遮挡内部 ；存在 false pass Place the spiral staircase inside a lighthouse tower interior: add a cylindrical ...	0129 • L6Q3 • neo-strip • 0.000 RF: 0005 • L6Q2 • ceiling-mounting-bracket • 0.000

GPT-5.6 Luna • observed rank #10

0.564 raw overall	181 成功出图	59 render failures	75.4% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- 低层：verifier 噪声与少量真实结构失败
 - L4：复杂局部结构与多条件满足不稳
 - L5/L6 主因：Blender API/类型错误，L6 仅 3/40 成功出图
 - 成功渲染后：主题语义和复杂场景 completeness
- RF exception class: TypeError 33; AttributeError 20; SyntaxError 4; NameError 1; KeyError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.750	0.750	30	0	10	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险：任务语义无效 + 独立归一化
L2	0.812	0.812	32	1	7	0	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险：内容被几何遮挡
L3	0.775	0.775	23	15	2	0	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险：对象重命名不可由像素验证
L4	0.625	0.625	23	4	13	0	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险：merge 拓扑不可由外观验证
L5	0.371	0.824	14	2	2	22	0011 • L5Q2 • curved-corner-bracket • 1.000 Redesign the bracket to feel lightweight and aerodynamic, like a component from a...	0095 • L5Q1 • helical-staircase • 0.000 RF: 0005 • L5Q1 • ceiling-mounting-bracket • 0.000
L6	0.051	0.675	0	3	0	37	—	0095 • L6Q0 • helical-staircase • 0.500 RF: 0005 • L6Q0 • ceiling-mounting-bracket • 0.000

Qwen3-Coder-Next • observed rank #11

0.472 raw overall	169 成功出图	71 render failures	70.4% 出图率
----------------------	-------------	-----------------------	--------------

失败类别

- L2-L4: Name/Syntax/Type、未定义变量及真实几何缺陷
- L5/L6 主因: Blender API/Key/Attribute/运行错误, 执行率最低
- 成功渲染后: 主题语义、可识别性和复杂场景 completeness
- 低层仍混入与全模型相同的视觉裁判误判

RF exception class: KeyError 23; AttributeError 16; NameError 7; TypeError 7; SyntaxError 5; RuntimeError 5; IndexError 3; ValueError 2; 其他/无日志 1; RecursionError 1; ImportError 1

Level	Score	Rendered mean	Full	Partial	Zero	RF	成功代表	非满分 / RF 代表
L1	0.800	0.800	32	0	8	0	0005 • L1Q0 • ceiling-mounting-bracket • 1.000 Change the bracket color from yellow to blue.	0060 • L1Q2 • mini-storage-drawers • 0.000 • 风险: 任务语义无效 + 独立归一化
L2	0.775	0.795	30	2	7	1	0005 • L2Q0 • ceiling-mounting-bracket • 1.000 Add a second mounting screw hole (with its boss) next to the existing one, positi...	0011 • L2Q3 • curved-corner-bracket • 0.000 • 风险: 内容被几何遮挡 RF: 0005 • L2Q2 • ceiling-mounting-bracket • 0.000
L3	0.683	0.719	16	19	3	2	0005 • L3Q1 • ceiling-mounting-bracket • 1.000 Modify the bracket in three ways: make the overall bracket body taller, reduce th...	0101 • L3Q3 • extrusion-2020 • 0.000 • 风险: 对象重命名不可由像素验证 RF: 0060 • L3Q1 • mini-storage-drawers • 0.000
L4	0.362	0.483	12	5	13	10	0129 • L4Q0 • neo-strip • 1.000 Add hexagonal void cutouts through the body of the strip channel at each pixel lo...	0095 • L4Q2 • helical-staircase • 0.000 • 风险: Boolean union 拓扑不可由外观验证 RF: 0011 • L4Q2 • curved-corner-bracket • 0.000 • 风险: merge 拓扑不可由外观验证
L5	0.125	0.500	3	6	1	30	0110 • L5Q1 • cam-profile • 1.000 Give the cam a futuristic sci-fi appearance, as if it were a precision component ...	0061 • L5Q2 • customizable-furniture-minis • 0.000 RF: 0005 • L5Q0 • ceiling-mounting-bracket • 0.000
L6	0.085	0.284	0	9	3	28	—	0129 • L6Q0 • neo-strip • 0.429 • 风险: 对象存在不等于关系/尺度成立 RF: 0005 • L6Q1 • ceiling-mounting-bracket • 0.000

5 · Sonnet 5 Verifier 专项审计

结论：有问题，而且不是边缘个案。已发现确定的同输入不同 verdict、稳定 false negative、任务—模态失配和高置信 false positive。Sonnet 5 可作初筛，不适合继续作为单裁 ground truth。

5.1 · 相同 RGBA 像素，不同分数

Case	像素相同组	判分矛盾	RGBA hash
0005 L1Q3	11/11	Sol/Luna=0; 其余 9=1	5fc91fb9b7
0011 L1Q3	11/11	Gemma=1; 其余 10=0	99511325e6
0129 L1Q2	11/11	3 个 0; 8 个 1	dd5be54179
0011 L2Q1	Sonnet 5 / Luna	0 vs 1	9cb7b546aa
0028 L2Q3	11/11	3 pass / 8 fail	b5aabd4a22
0095 L2Q1	11/11	Gemma=.5; 其余 10=1	bff26526f6
0110 L2Q2	5 个模型	仅 Sol=1; 其余 4=0	a1995db75f

hash 按六张解码后 RGBA 像素串联计算。Evaluator 消息不包含被测模型名，因此差异只能来自 judge 调用/解析非确定性或协议不稳定。

5.2 · 5×复测与高置信 false negative

复测项	结果	独立证据
blue control · 0005 L1Q0	5/5 score 1	颜色变化清楚
sofa height · 0061 L1Q1	5/5 score 0	front bbox 高度 338→437 px
sofa length · 0061 L1Q2	5/5 score 0	right aspect 1.448→2.031
stair width · 0095 L1Q3	5/5 score 0	top bbox 176→236 px
Boolean union · 0095 L4Q2	5/5 score 0	代码执行 UNION/EXACT 并删除 operand

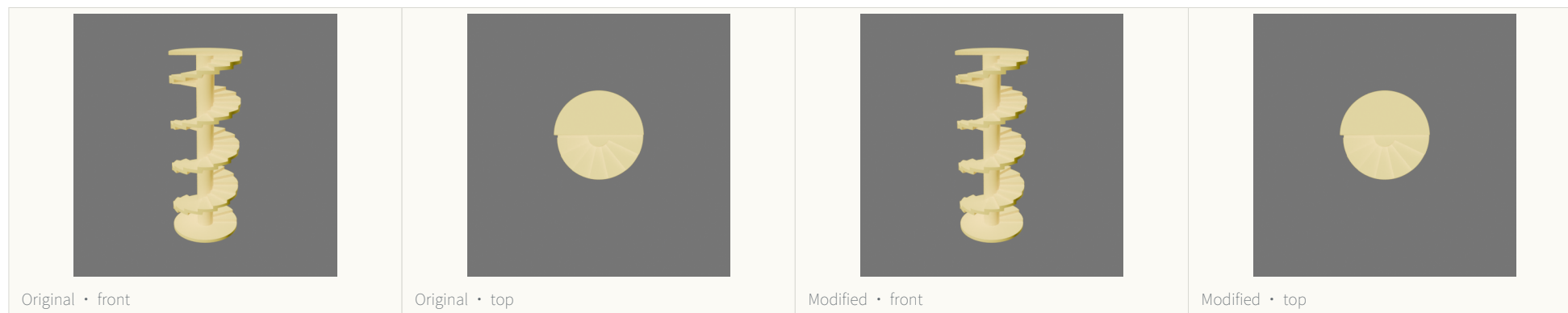


Sofa 高度与长度的前后对照诊断图。

5.3 · 任务—模态失配

- L3 rename: 11 个脚本都将对象改名为 aluminum_profile_2020; RGB 六视图没有 object metadata, 第三项 11/11 判 no。
- L4 topology: Terra 对 base 与 tread 做 EXACT Boolean UNION; 外观相近不能证明 mesh connectivity。
- 独立 normalize: ORIGINAL/MODIFIED 分别 center + scale=1/max_dim, 削弱整体尺寸并抹掉平移。
- invalid task: 0060 L1Q2 把 DEPTH 描述为前后深度, 但源脚本 alias 到 WALL_HEIGHT, 实际控制 Z 高度。
- 小型内部 feature: pocket、relief、keyway、column 被遮挡或低于可靠像素尺度, judge 不应默认 no。

Terra Boolean union: 像素不能证明 topology



5.4 • False-positive 分层抽样

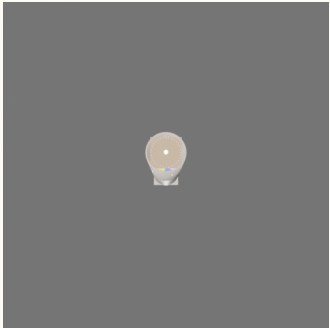
模型 / case	Sonnet 5	独立复核	判满失败在哪里
Gemma • 0028 L6Q0	7/7	4/7	封闭圆筒不读作 interior；无可辨水平 railing；lantern 只是圆柱
Haiku • 0129 L6Q0	7/7	2/7	channel 悬出桌面；iron/stand 分离；cutters 不可辨；无 desk-lamp light
Qwen • 0028 L5Q0	4/4	3/4	干净纯色金属没有 aged/weathered 证据；bottom view 扶手尺度异常
Gemma • 0129 L5Q3	3/3	疑似 2/3	结构与中性色成立，但 premium/luxury 视觉证据偏弱

抽样由 11 个模型的分层核心样本 + 5 个高风险风格样本组成，不能把 3/16 外推为 228 个 L5/L6 full pass 的总体误判率。

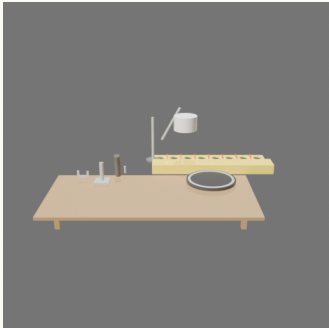
False-positive 视觉证据



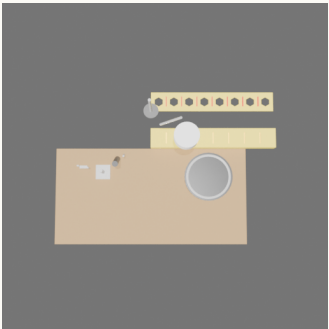
Modified • front



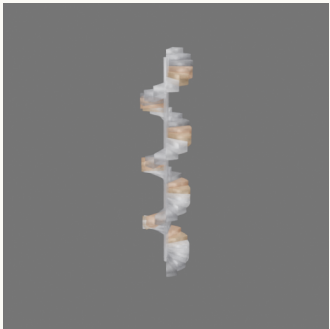
Modified • top



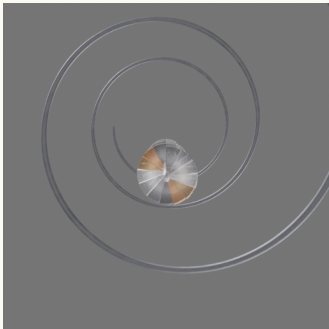
Modified • front



Modified • top



Modified • front



Modified • bottom

5.5 · 调用、解析与可追溯性

问题	已发生事实 / 风险	建议
84 个首次空/不可解析	57 次重新调用；27 次从原 raw 重解析；最终 schema 已修复	报告 rerun rate；保存首次与重跑 lineage
无 rationale/confidence	多数正式 eval 只存 yes/no 与 score	保存证据 view、rationale、confidence、abstain
parser 校验不足	主 evaluator 未严格校验长度、item identity、多余答案	schema + item identity + exact answer count
有效配置不可复现	结果目录未固化 request 参数、seed、代码 commit/dirty diff	落盘 resolved model/config/prompt/code hash
角色重叠	Sonnet 5 生成 query、被测、又做 verifier；未发现明确自偏爱	盲化输入、跨家族二裁、majority/abstain

5.6 · 推荐替代协议

Level	主验证器	图像角色	发布指标
L1	AST/literal + scene invariant + bbox	只检查渲染退化与目标方向	parameter success / visual sanity
L2/L3	对象数、尺寸、metadata、组件关系	close-up / section view	geometry + observability
L4	mesh connectivity、boolean、through-hole、watertight	只验证外观结果	topology + visual pass 分开
L5/L6	盲化跨家族多裁判；confidence/abstain	共享 camera；interior/cutaway views	raw、出图率、rendered-only、二裁率

6 · 执行失败与模型稳定性矩阵

模型	总 RF	主要异常类型	L1	L2	L3	L4	L5	L6	结论
Claude Opus 4.8	8	AttributeError 2, 其他/无日志 2, TypeError 2, SyntaxError 1	0	0	0	0	5	3	L6: 道具 completeness、遮挡、尺度与场景连贯性
Claude Sonnet 4.6	26	KeyError 10, TypeError 9, AttributeError 5, SyntaxError 1	0	1	2	1	10	12	L6: 复杂场景缺件、尺度、遮挡与连贯性
Claude Sonnet 5	3	AttributeError 2, TypeError 1	0	0	0	0	1	2	L6: 复杂场景 completeness 和内容可见性
Claude Haiku 4.5	29	AttributeError 12, TypeError 9, KeyError 4, RuntimeError 3	0	0	0	1	12	16	L6: 道具缺失、遮挡、尺度与场景连贯性
GPT-5.6 Sol	56	AttributeError 34, TypeError 21, KeyError 1	0	0	0	4	20	32	成功渲染后：风格细化、道具数量与整体语义偶有不足
GPT-5.6 Terra	53	AttributeError 28, TypeError 25	0	0	0	0	19	34	成功渲染后：有机化、场景配件和可见性不足
GPT-5.6 Luna	59	TypeError 33, AttributeError 20, SyntaxError 4, NameError 1	0	0	0	0	22	37	成功渲染后：主题语义和复杂场景 completeness
GPT-5.5	36	AttributeError 32, TypeError 4	0	0	0	1	19	16	复杂场景：遮挡、道具 completeness 与视觉证据不足
gpt-oss-120b	27	KeyError 11, AttributeError 6, TypeError 4, NameError 2	0	1	0	0	12	14	L5/L6: 风格语义、场景道具、尺度和连贯性
Gemma 4 31B	34	TypeError 14, AttributeError 13, KeyError 3, 其他/无日志 1	0	0	0	3	16	15	L6: 场景遮挡；另有唯一 full pass 为高置信 false positive
Qwen3-Coder-Next	71	KeyError 23, AttributeError 16, NameError 7, TypeError 7	0	1	2	10	30	28	低层仍混入与全模型相同的视觉裁判误判

解释：同一个 score=0 可能是完全不同的机制。RF 是代码/API 稳定性；rendered zero 可能是真实几何/语义失败，也可能是不可观察或 verifier error。榜单必须同时发布 raw mean、出图率和 rendered-only mean。

附录 A • 60 个任务的 observed 难度

每行聚合 11 个模型。风险标签来自脚本/像素/重复判定专项审计。

Level 1 任务矩阵

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0060 • L1Q2 • mini-storage-drawers	Increase the depth (DEPTH) of the cabinet from 68.0 to 90.0.	0.000	0	0	11	0	任务语义无效 + 独立归一化
0060 • L1Q3 • mini-storage-drawers	Change the bottom rail thickness (BOTTOM_RAIL_THICKNESS) from 1.0 to 3.0.	0.000	0	0	11	0	目标被外壳遮挡
0061 • L1Q1 • customizable-furniture-minis	Increase the armchair's overall height from 34 to 50.	0.000	0	0	11	0	高置信 false negative
0061 • L1Q2 • customizable-furniture-minis	Increase the sofa length from 54 to 80.	0.000	0	0	11	0	高置信 false negative
0095 • L1Q3 • helical-staircase	Increase the step_width parameter from 24.0 to 35.0 to make the treads extend further outward.	0.000	0	0	11	0	高置信 false negative
0101 • L1Q2 • extrusion-2020	Increase the core bore diameter from 4.2mm to 6mm.	0.000	0	0	11	0	高置信 false negative
0101 • L1Q3 • extrusion-2020	Reduce the corner radius of the profile from 0.5 to 0.1 to make the corners sharper.	0.000	0	0	11	0	sub-pixel, 不宜视觉验证
0011 • L1Q3 • curved-corner-bracket	Reduce the bracket thickness from 2.5 to 1.0.	0.091	1	0	10	0	确定 verifier 误判
0129 • L1Q2 • neo-strip	Increase the hex_radius parameter from 5.0 to 7.0.	0.727	8	0	3	0	同像素判分翻转
0005 • L1Q3 • ceiling-mounting-bracket	Increase the standoff height from 15.0 to 25.0.	0.818	9	0	2	0	同像素判分翻转
0005 • L1Q0 • ceiling-mounting-bracket	Change the bracket color from yellow to blue.	1.000	11	0	0	0	—
0005 • L1Q1 • ceiling-mounting-bracket	Increase the mounting screw radius from 2.5 to 4.0.	1.000	11	0	0	0	—
0005 • L1Q2 • ceiling-mounting-bracket	Reduce the wall thickness from 2.0 to 1.0.	1.000	11	0	0	0	—
0011 • L1Q0 • curved-corner-bracket	Change the bracket color from yellow to blue.	1.000	11	0	0	0	—
0011 • L1Q1 • curved-corner-bracket	Increase the bracket width from 10.0 to 20.0.	1.000	11	0	0	0	—
0011 • L1Q2 • curved-corner-bracket	Increase the number of holes from 3 to 5.	1.000	11	0	0	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0028 • L1Q0 • parametric-spiral-stairs	Change the red step color to black.	1.000	11	0	0	0	—
0028 • L1Q1 • parametric-spiral-stairs	Increase the central column radius from 10.0 to 25.0.	1.000	11	0	0	0	—
0028 • L1Q2 • parametric-spiral-stairs	Change the staircase radius from 100.0 to 150.0.	1.000	11	0	0	0	—
0028 • L1Q3 • parametric-spiral-stairs	Change the silver column material color to gold.	1.000	11	0	0	0	—
0060 • L1Q0 • mini-storage-drawers	Change the color of the storage drawer cabinet from gold to red.	1.000	11	0	0	0	—
0060 • L1Q1 • mini-storage-drawers	Increase the wall thickness (WALL) from 1.0 to 2.0.	1.000	11	0	0	0	—
0061 • L1Q0 • customizable-furniture-minis	Change the armchair color from yellow to red.	1.000	11	0	0	0	—
0061 • L1Q3 • customizable-furniture-minis	Reduce the material roughness value from 0.58 to 0.1.	1.000	11	0	0	0	—
0064 • L1Q0 • tool-shelf-for-bench	Change the shelf color from yellow to red.	1.000	11	0	0	0	—
0064 • L1Q1 • tool-shelf-for-bench	Increase the wall thickness from 3.0 to 6.0.	1.000	11	0	0	0	—
0064 • L1Q2 • tool-shelf-for-bench	Reduce the number of holes per shelf level from 12 to 8.	1.000	11	0	0	0	—
0064 • L1Q3 • tool-shelf-for-bench	Increase the rear shelf height from 100.0 to 130.0.	1.000	11	0	0	0	—
0095 • L1Q0 • helical-staircase	Change the staircase color from yellow to blue.	1.000	11	0	0	0	—
0095 • L1Q1 • helical-staircase	Increase the core_radius parameter from 8.0 to 14.0 so the central column is thicker.	1.000	11	0	0	0	—
0095 • L1Q2 • helical-staircase	Reduce the number of revolutions from 4 to 2.	1.000	11	0	0	0	—
0101 • L1Q0 • extrusion-2020	Change the aluminum extrusion color from gold to silver gray.	1.000	11	0	0	0	—
0101 • L1Q1 • extrusion-2020	Increase the extrusion length from 60mm to 100mm.	1.000	11	0	0	0	—
0110 • L1Q0 • cam-profile	Change the cam disk color from yellow to blue.	1.000	11	0	0	0	—
0110 • L1Q1 • cam-profile	Increase the cam thickness from 8.0 to 14.0.	1.000	11	0	0	0	—
0110 • L1Q2 • cam-profile	Increase the shaft hole diameter from 6.0 to 10.0.	1.000	11	0	0	0	—
0110 • L1Q3 • cam-profile	Increase the hub extra height from 3.0 to 6.0.	1.000	11	0	0	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0129 • L1Q0 • neo-strip	Change the body color of the strip channel from yellow to blue.	1.000	11	0	0	0	—
0129 • L1Q1 • neo-strip	Change the color of the light-blocking dividers from red to green.	1.000	11	0	0	0	—
0129 • L1Q3 • neo-strip	Increase the body_height from 10.0 to 15.0.	1.000	11	0	0	0	—

Level 2 任务矩阵

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0011 • L2Q3 • curved-corner-bracket	Remove the corner fill block so the two straight legs meet with an open gap at the inner corner instead of being joined by a filled block.	0.000	0	0	11	0	内容被几何遮挡
0028 • L2Q0 • parametric-spiral-stairs	Replace the central column with a rectangular box pillar instead of a cylinder.	0.000	0	0	11	0	中心柱被踏步遮挡
0101 • L2Q3 • extrusion-2020	Replace the four thin rectangular diagonal relief channels with small circular holes instead.	0.000	0	0	10	1	细节不可可靠视觉辨认
0028 • L2Q2 • parametric-spiral-stairs	Change the tread shape so each step is a full rectangular slab spanning the entire radius width instead of a pie-slice sector.	0.091	1	0	10	0	—
0028 • L2Q3 • parametric-spiral-stairs	Make the staircase spiral in the opposite rotational direction (reverse the winding direction).	0.273	3	0	8	0	11 模型同像素、3/8 分裂
0005 • L2Q3 • ceiling-mounting-bracket	Reorient the mounting screw hole and boss so the hole axis points vertically (up/down through the top) instead of horizontally through the side wall.	0.364	4	0	7	0	—
0011 • L2Q1 • curved-corner-bracket	Reduce the number of large holes along the curved hoop from 3 to 1, placed at the center of the arc.	0.455	5	0	6	0	同像素相反 verdict
0129 • L2Q3 • neo-strip	Enable the interior hexagonal voids inside the main channel body so each pixel position has a hexagonal cutout through the body, not just the cover plate.	0.455	5	0	6	0	—
0101 • L2Q1 • extrusion-2020	Replace the four cylindrical internal pockets with square pockets instead.	0.500	0	11	0	0	细节不可可靠视觉辨认
0110 • L2Q2 • cam-profile	Rotate the keyway slot in the central shaft hole so it opens horizontally (left-right) instead of vertically (up/down).	0.545	6	0	5	0	同像素相反 verdict
0060 • L2Q3 • mini-storage-drawers	Increase the number of narrow column compartments from 4 to 5, adding an extra half-height and normal-height drawer slot.	0.727	8	0	3	0	—
0005 • L2Q2 • ceiling-mounting-bracket	Increase the number of gripping fingers that hold the mounted part from 2 to 3, evenly spaced across the bracket's width.	0.773	8	1	0	2	—
0061 • L2Q0 • customizable-furniture-minis	Replace the two rectangular armrests with cylindrical (round) armrests.	0.864	9	1	1	0	—
0110 • L2Q1 • cam-profile	Modify the cam lift profile so it has two lift lobes (two rise-dwell-fall cycles) evenly spaced around the disk instead of just one.	0.909	10	0	1	0	—
0005 • L2Q1 • ceiling-mounting-bracket	Change the mounting screw boss and its hole from a cylindrical shape to a rectangular (square) block with a square hole.	0.955	10	1	0	0	—
0095 • L2Q1 • helical-staircase	Reduce the number of revolutions of the staircase from 4 to 2.	0.955	10	1	0	0	11 模型同像素判分翻转
0005 • L2Q0 • ceiling-mounting-bracket	Add a second mounting screw hole (with its boss) next to the existing one, positioned side by side along the width of the bracket's rear wall.	1.000	11	0	0	0	—
0011 • L2Q0 • curved-corner-bracket	Increase the number of small mounting holes along each straight leg from 3 to 5.	1.000	11	0	0	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0011 • L2Q2 • curved-corner-bracket	Make the circular segment resolution much lower so the curved hoop looks faceted/polygonal instead of smooth.	1.000	11	0	0	0	—
0028 • L2Q1 • parametric-spiral-stairs	Reduce the number of steps per full turn from 13 to 8, making each tread cover a wider angle.	1.000	11	0	0	0	—
0060 • L2Q0 • mini-storage-drawers	Reduce the number of narrow column compartments (the half-height and normal-height drawers) from 4 to 3.	1.000	11	0	0	0	—
0060 • L2Q1 • mini-storage-drawers	Split the single quad-width compartment in the top row into two separate double-width compartments side by side.	1.000	11	0	0	0	—
0060 • L2Q2 • mini-storage-drawers	Merge the two double-width compartments in their row into a single wide compartment spanning the whole row.	1.000	11	0	0	0	—
0061 • L2Q1 • customizable-furniture-minis	Change the armchair so only one armrest exists, on the left side, removing the right armrest entirely.	1.000	11	0	0	0	—
0061 • L2Q2 • customizable-furniture-minis	Make the backrest only half the width of the seat instead of spanning the full width, positioned on the left side.	1.000	11	0	0	0	—
0061 • L2Q3 • customizable-furniture-minis	Split the single seat cushion into two separate square cushions placed side by side instead of one continuous cushion.	1.000	11	0	0	0	—
0064 • L2Q0 • tool-shelf-for-bench	Reduce the number of horizontal back slats on the rear panel from 4 to 2, keeping them evenly spaced with visible gaps between them.	1.000	11	0	0	0	—
0064 • L2Q1 • tool-shelf-for-bench	Change the mounting holes on all three shelf levels from circular holes to square holes.	1.000	11	0	0	0	—
0064 • L2Q2 • tool-shelf-for-bench	Remove the side walls from the middle shelf level, leaving only the rear and front shelves with side walls.	1.000	11	0	0	0	—
0064 • L2Q3 • tool-shelf-for-bench	Add a second center support wall so there are two evenly spaced support walls across the shelf width instead of just one.	1.000	11	0	0	0	—
0095 • L2Q0 • helical-staircase	Change the top landing from a half-disc (180 degrees) to a full circular disc (360 degrees).	1.000	11	0	0	0	—
0095 • L2Q2 • helical-staircase	Replace the solid cylindrical base disk at the bottom with a sector-shaped base matching the half-disc landing style instead of a full circle.	1.000	11	0	0	0	—
0095 • L2Q3 • helical-staircase	Increase the number of steps per revolution from 14 to 20, making the spiral turns denser with more, shorter steps per loop.	1.000	11	0	0	0	—
0101 • L2Q0 • extrusion-2020	Reduce the T-slots from 4 sides to only 2 opposite sides, leaving the other two sides flat without any slot cutout.	1.000	11	0	0	0	—
0101 • L2Q2 • extrusion-2020	Change the central bore hole from a circular cylinder to a square hole.	1.000	11	0	0	0	—
0110 • L2Q0 • cam-profile	Change the raised hub boss around the shaft hole from a smooth circular cylinder into a hexagonal prism.	1.000	11	0	0	0	—
0110 • L2Q3 • cam-profile	Replace the circular shaft hole with a hexagonal shaft hole, keeping the keyway cut into it.	1.000	11	0	0	0	—
0129 • L2Q0 • neo-strip	Reduce the number of pixel slots from 8 to 5.	1.000	11	0	0	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0129 • L2Q1 • neo-strip	Replace the hexagonal pixel holes in the cover plate with round circular holes.	1.000	11	0	0	0	—
0129 • L2Q2 • neo-strip	Add divider slots at both the start and end of the channel so every pixel is fully separated, including the first and last ones.	1.000	11	0	0	0	—

Level 3 任务矩阵

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0101 • L3Q3 • extrusion-2020	Remove the diagonal relief cutters entirely, shrink the internal pocket holes to a smaller diameter, and rename the object to 'aluminum_profile_2020'.	0.061	0	2	9	0	对象重命名不可由像素验证
0110 • L3Q3 • cam-profile	Apply these three edits: shift the fall_start and fall_end angles so the dwell (flat) region after the lift is shorter, reduce base_circle_radius to make the overall disk smaller, and change the material to a metallic look with high metallic value.	0.182	0	5	6	0	—
0095 • L3Q3 • helical-staircase	Perform three separate modifications: double the overall number of steps (step_count) to make the staircase taller, make the landing sweep a full 360 degrees instead of 180, and add a bevel/edge highlight line material in a contrasting red color.	0.333	0	11	0	0	—
0028 • L3Q1 • parametric-spiral-stairs	Make the central column thicker, increase the step height so the stairs are taller and more spaced out, and change the color palette to shades of blue instead of rainbow colors.	0.364	0	11	0	0	全模型正确参数、系统性漏判
0101 • L3Q0 • extrusion-2020	Make three independent changes: increase the extrusion length to 100mm, enlarge the center bore diameter to 6mm, and change the aluminum color to silver/gray instead of gold.	0.364	0	11	0	0	全模型正确参数、系统性漏判
0005 • L3Q2 • ceiling-mounting-bracket	Update the model with these independent changes: make the rear screw boss faceted (low-poly, like an octagon) instead of smoothly round, widen the overall bracket to accommodate a wider mounted object, and change the material to a metallic gray color.	0.485	0	11	0	0	—
0060 • L3Q3 • mini-storage-drawers	Perform these independent edits: increase the horizontal spacing between the narrow columns, reduce the depth of the double-width compartments, and color the bottom rails a different color from the rest of the cabinet.	0.515	2	7	2	0	—
0061 • L3Q0 • customizable-furniture-minis	Make the backrest taller, change the seat cushion color to blue, and reduce the arm rest width so the seat area looks wider.	0.545	1	10	0	0	—
0101 • L3Q2 • extrusion-2020	Increase the wall thickness to make the walls sturdier, double the number of circle segments for smoother rounded corners, and change the material to a matte black finish.	0.576	4	6	1	0	—
0095 • L3Q0 • helical-staircase	Make three independent changes: reduce the number of revolutions from 4 to 2, increase the step_depth so each tread is thicker, and change the staircase material color from yellow to blue.	0.606	0	11	0	0	—
0095 • L3Q2 • helical-staircase	Apply these independent edits: increase steps_per_revolution so there are more, narrower steps per full turn, change the base disk's material to a dark gray, and increase the overlap_angle so consecutive treads overlap more.	0.606	4	7	0	0	—
0060 • L3Q2 • mini-storage-drawers	Make the following independent changes: reduce the number of narrow columns from four to three, increase the height of the half-height row of compartments, and give the surface a glossier, shinier finish.	0.667	2	8	0	1	—
0129 • L3Q1 • neo-strip	Change the divider color from red to white, add the interior hexagonal voids to the main body (include_interior_hexes), and increase the cover plate thickness so it appears noticeably thicker.	0.667	1	10	0	0	—
0028 • L3Q2 • parametric-spiral-stairs	Reduce the number of steps per turn to make each tread cover a wider angle, shrink the column radius to make it thinner, and add a slight metallic silver tint change to gold for the column.	0.697	4	6	0	1	—
0110 • L3Q0 • cam-profile	Make three independent changes: increase the max_lift value so the cam profile is noticeably more eccentric/oval, double the cam_thickness to make the plate thicker, and change the material color from yellow to blue.	0.697	3	8	0	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0129 • L3Q0 • neo-strip	Increase the pixel count from 8 to 12, change the body color from yellow to blue, and increase the body height so the strip is taller.	0.697	1	10	0	0	全模型正确高度、10/11 漏判
0061 • L3Q1 • customizable-furniture-minis	Increase the overall height of the armchair, make the left arm rest taller than the right arm rest, and change the material color of the backrest to red.	0.727	2	9	0	0	—
0060 • L3Q0 • mini-storage-drawers	Make three independent changes: recolor the entire cabinet from gold to blue, increase the overall depth (front-to-back thickness) of the cabinet by about 50%, and make the bottom support rails noticeably thicker than they currently are.	0.758	3	8	0	0	—
0011 • L3Q2 • curved-corner-bracket	Apply the following changes: extend the overall length of the bracket's legs, make the curved corner more angular/faceted by reducing the number of curve segments, and change the bracket color to green.	0.788	5	6	0	0	—
0011 • L3Q3 • curved-corner-bracket	Modify the bracket with three separate changes: enlarge the diameter of the small mounting holes, remove the large holes positioned along the outer curved section, and change the bracket's material color to orange.	0.818	6	5	0	0	—
0061 • L3Q3 • customizable-furniture-minis	Make the backrest shorter in length (narrower along the width), increase the height of both arm rests, and change the seat cushion to a glossy dark brown color.	0.818	7	3	0	1	—
0064 • L3Q1 • tool-shelf-for-bench	Make the front shelf's hole diameter larger than the rear and mid holes, shorten the front_height so the front tier is lower, and remove the rear back slats entirely.	0.818	7	3	1	0	—
0005 • L3Q3 • ceiling-mounting-bracket	Apply three separate edits: thin down the wall thickness of the bracket, enlarge the mounting screw hole radius, and make the mounting slot for the mounted object shallower (reduce its height).	0.848	7	4	0	0	—
0061 • L3Q2 • customizable-furniture-minis	Shrink the seat cushion thickness, add more rounding (larger radius) to the arm rests, and change the arm rest color to green while keeping the rest of the chair yellow.	0.848	6	5	0	0	—
0064 • L3Q2 • tool-shelf-for-bench	Increase the shelf_depth_front so the bottom tier extends further forward, decrease the number_of_holes so the shelf is narrower overall, and change the material color to a dark gray instead of yellow.	0.848	6	5	0	0	—
0064 • L3Q3 • tool-shelf-for-bench	Add center support walls even when the shelf is narrow (force them to always appear regardless of width), increase the spacing_of_holes to spread the holes further apart, and make the mid shelf's depth shallower than before.	0.879	8	3	0	0	—
0060 • L3Q1 • mini-storage-drawers	Apply three separate edits: shrink the width of the large quad-width compartment at the top row, increase the thickness of all the internal divider walls so they look chunkier, and change the material color to matte red.	0.909	10	0	0	1	—
0095 • L3Q1 • helical-staircase	Modify the staircase in three ways: widen the step_width so the treads extend further outward, shrink the core_radius so the central column is thinner, and remove the top landing disc entirely.	0.939	9	2	0	0	—
0005 • L3Q0 • ceiling-mounting-bracket	Make three independent changes to the bracket: increase the mounting screw hole radius from 2.5mm to about 5mm, increase the wall thickness around the slot from 2mm to about 4mm, and change the material color from yellow to green.	0.970	10	1	0	0	—
0110 • L3Q2 • cam-profile	Make the following changes: reduce roller_radius so the cam lobe has a sharper, more pronounced peak, increase hub_extra_dia so the hub boss is wider, and set the material roughness lower so the surface looks glossier/shinier.	0.970	10	1	0	0	—
0005 • L3Q1 • ceiling-mounting-bracket	Modify the bracket in three ways: make the overall bracket body taller, reduce the standoff height so there is less clearance above the mounting slot, and shorten the depth of the side finger tabs that grip the mounted object.	1.000	11	0	0	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0011 • L3Q0 • curved-corner-bracket	Make three independent changes to the bracket: increase the number of small mounting holes from 3 to 5, completely remove the large curved holes along the outer arc, and change the bracket's color from yellow to blue.	1.000	11	0	0	0	—
0011 • L3Q1 • curved-corner-bracket	Update the bracket with these independent edits: increase the width (depth along the extrusion axis) so the bracket looks thicker front-to-back, increase the wall thickness of the straight legs, and recolor the whole bracket to matte gray.	1.000	11	0	0	0	—
0028 • L3Q0 • parametric-spiral-stairs	Increase the staircase radius so the treads are wider, reduce the number of revolutions to 2, and change the central column color from silver to black.	1.000	11	0	0	0	—
0028 • L3Q3 • parametric-spiral-stairs	Double the number of revolutions for a taller spiral, decrease the step height so treads are more tightly packed, and change the tread colors to a two-tone black-and-white alternating pattern.	1.000	11	0	0	0	—
0064 • L3Q0 • tool-shelf-for-bench	Reduce the number of holes per shelf from 12 to 8, increase the wall_thickness from 3.0 to 6.0, and change the rear_height from 100.0 to 130.0.	1.000	11	0	0	0	—
0101 • L3Q1 • extrusion-2020	Change the profile size from 20x20mm to 30x30mm, reduce the corner radius to make sharper corners, and make the slot width narrower.	1.000	11	0	0	0	—
0110 • L3Q1 • cam-profile	Update the model with these changes: increase the shaft_hole_dia so the center hole is larger, remove the keyway notch entirely so the hole is a plain circle, and increase the hub_extra_height so the raised hub boss is taller.	1.000	11	0	0	0	—
0129 • L3Q2 • neo-strip	Make the hexagonal pixel holes larger (increase hex_radius), reduce the pixel spacing so the pixels are closer together, and enable the start half slot so there's an extra divider at the beginning of the strip.	1.000	11	0	0	0	—
0129 • L3Q3 • neo-strip	Widen the body (body_width) so the strip is noticeably wider, shrink the divider thickness so the dividers are thinner, and change the strip channel void material tint by recoloring the main body to green.	1.000	11	0	0	0	—

Level 4 任务矩阵

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0011 • L4Q2 • curved-corner-bracket	Merge the two straight legs and the corner fill block into a single continuous L-shaped piece by rounding the inner corner where they meet, blending it smoothly with the curved outer hoop.	0.000	0	0	10	1	merge 拓扑不可由外观验证
0028 • L4Q1 • parametric-spiral-stairs	Cut a circular hole through the center of every tread so the central column passes cleanly through each step without any overlapping geometry.	0.000	0	0	10	1	—
0095 • L4Q2 • helical-staircase	Merge the base disk and the bottom-most tread into a single continuous solid piece so there is no visible seam between them.	0.000	0	0	11	0	Boolean union 拓扑不可由外观验证
0028 • L4Q3 • parametric-spiral-stairs	Merge the central column and all the treads into a single continuous solid piece.	0.091	1	0	8	2	—
0064 • L4Q3 • tool-shelf-for-bench	Add a raised front lip along the top front edge of the front-level shelf plate to keep small tools from rolling off, connecting seamlessly to the existing left and right side walls at that level.	0.091	1	0	10	0	—
0110 • L4Q0 • cam-profile	Cut a radial set-screw hole through the wall of the raised hub boss that breaks through into the central shaft bore, so a screw could clamp onto a shaft.	0.091	1	0	8	2	—
0129 • L4Q3 • neo-strip	Merge the divider walls into the cover plate so they form one continuous solid piece with no visible seam, instead of separate divider blocks sitting on the cover.	0.091	1	0	10	0	—
0061 • L4Q3 • customizable-furniture-minis	Add a horizontal support bar connecting the bottom of the left and right armrests underneath the seat cushion.	0.182	2	0	8	1	—
0028 • L4Q0 • parametric-spiral-stairs	Add a helical handrail that spirals along the outer edge of the treads, following the staircase from bottom to top.	0.364	2	3	4	2	—
0061 • L4Q2 • customizable-furniture-minis	Merge the two arm rests and the backrest into a single continuous piece of upholstery that wraps around the seat without visible seams.	0.364	4	0	7	0	—
0005 • L4Q2 • ceiling-mounting-bracket	Add a triangular gusset rib connecting the vertical screw boss to the main bracket body for extra support.	0.455	5	0	5	1	—
0005 • L4Q3 • ceiling-mounting-bracket	Merge the two side finger tabs with the main body using filleted (rounded) transitions instead of the current sharp step cuts.	0.455	2	8	1	0	—
0028 • L4Q2 • parametric-spiral-stairs	Add support brackets connecting each tread to the central column so the treads look structurally attached instead of just touching the column.	0.455	5	0	6	0	—
0005 • L4Q0 • ceiling-mounting-bracket	Add a second mounting screw boss with a through-hole on the opposite side wall of the bracket, mirroring the existing rear screw boss.	0.500	1	10	0	0	—
0005 • L4Q1 • ceiling-mounting-bracket	Cut two additional small mounting screw holes through the base flanges of the bracket, one on each side of the slot opening.	0.545	4	3	4	0	—
0064 • L4Q2 • tool-shelf-for-bench	Cut a rectangular pegboard-style opening through the rear slats (between the existing slat bars) to allow long-handled tools to hang through the back of the shelf.	0.545	6	0	5	0	—
0101 • L4Q3 • extrusion-2020	Cut a small threaded mounting hole through one flat outer face of the extrusion, positioned so it breaks through into the nearest internal pocket, allowing a screw to reach the internal cavity from outside.	0.545	5	3	3	0	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0129 • L4Q2 • neo-strip	Add small alignment pegs on the underside of the cover plate that fit into corresponding holes cut into the top of the strip channel body, so the cover plate locks into place when assembled.	0.576	5	3	3	0	—
0011 • L4Q1 • curved-corner-bracket	Cut a triangular notch out of the inner curved edge of the bracket, positioned at the midpoint of the curve, so the arc is interrupted by an open gap.	0.591	5	2	4	0	—
0064 • L4Q0 • tool-shelf-for-bench	Cut two bolt-mounting holes through the bottom of each side wall (left and right) so the shelf can be bolted directly onto the workbench surface.	0.606	6	1	4	0	—
0095 • L4Q3 • helical-staircase	Cut a rectangular archway opening through the base disk on one side to allow walking underneath the staircase.	0.606	6	1	3	1	—
0101 • L4Q1 • extrusion-2020	Add a flush countersink (conical chamfer) at both ends of the central bore so a bolt head can sit flush with the extrusion's end faces.	0.606	6	1	4	0	—
0095 • L4Q1 • helical-staircase	Add a spiral handrail that follows the outer edge of the treads, supported by vertical balusters connecting each tread to the rail.	0.636	4	6	1	0	—
0061 • L4Q0 • customizable-furniture-minis	Cut a row of decorative holes through the backrest so they pass all the way through from front to back.	0.667	5	4	0	2	—
0011 • L4Q0 • curved-corner-bracket	Add a fourth mounting hole through the curved arm, positioned between the outermost existing hole and the tip of the corner, so it passes fully through the material like the others.	0.727	7	4	0	0	—
0011 • L4Q3 • curved-corner-bracket	Add a small reinforcement gusset (triangular wall) connecting the underside of the curved hoop to the top surface of the straight legs, positioned near the bracket's corner.	0.727	8	0	3	0	—
0060 • L4Q0 • mini-storage-drawers	Cut a circular cable-management hole through the back wall of the top quad-width compartment so it connects directly to that compartment's interior.	0.727	7	2	0	2	—
0060 • L4Q1 • mini-storage-drawers	Add a small rectangular pull handle to the front face of each of the four narrow half-height drawer compartments, attached flush to the front wall.	0.727	8	0	3	0	—
0095 • L4Q0 • helical-staircase	Cut a cylindrical hole through the base disk and landing so the central core column becomes a hollow shaft passing through the entire staircase.	0.818	9	0	2	0	—
0101 • L4Q2 • extrusion-2020	Seal one end of the extrusion with a solid end cap plate that covers the T-slots, bore, and internal pockets, fused as a continuous part of the body.	0.818	9	0	1	1	—
0061 • L4Q1 • customizable-furniture-minis	Add wooden legs attached to the bottom of the armchair so it appears raised off the ground.	0.848	8	2	1	0	—
0110 • L4Q1 • cam-profile	Add a ring of evenly spaced weight-reduction holes through the flat body of the cam plate, positioned between the hub and the outer cam profile edge so they don't intersect either feature.	0.848	9	1	0	1	—
0101 • L4Q0 • extrusion-2020	Drill a cross hole through the extrusion near one end, perpendicular to the extrusion's length, so it passes through both opposite side walls and intersects the central bore, allowing a connector pin to pass through the whole profile.	0.879	9	1	1	0	—
0060 • L4Q2 • mini-storage-drawers	Add a vertical divider wall inside the quad-width top compartment to split it into two equal side-by-side openings, merging seamlessly with the existing wall grid structure.	0.909	10	0	0	1	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0060 • L4Q3 • mini-storage-drawers	Merge all the bottom rails into a single continuous base plate that spans the full width and depth of the cabinet, structurally connected to the bottom edge of the outer shell.	0.909	10	0	0	1	visual pass 不证明 topology
0064 • L4Q1 • tool-shelf-for-bench	Merge the three separate center support walls (rear, mid, front) into a single continuous center divider panel that spans from the base up to the top of the rear shelf, integrated with all three shelf plates it passes through.	0.909	9	2	0	0	—
0110 • L4Q2 • cam-profile	Add a circular counterbore recess on the bottom face of the cam plate, concentric with the central shaft hole, to seat a bearing flush with the surface.	0.909	10	0	0	1	—
0129 • L4Q1 • neo-strip	Cut mounting screw holes through the base of the strip channel body, positioned between the divider slots, so the strip can be screwed onto a surface.	0.909	10	0	1	0	—
0110 • L4Q3 • cam-profile	Cut a small flat notch into the outer rim of the cam profile at the point of maximum lift to act as a timing index mark, blending cleanly into the existing cam surface.	0.939	10	1	0	0	—
0129 • L4Q0 • neo-strip	Add hexagonal void cutouts through the body of the strip channel at each pixel location, matching the hexagonal holes already present in the cover plate, so the light can pass through the main body as well as the cover.	1.000	11	0	0	0	—

Level 5 任务矩阵

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0095 • L5Q1 • helical-staircase	Transform the staircase into a medieval castle tower stairway, giving it a heavy stone-fortress aesthetic.	0.152	0	3	4	4	—
0095 • L5Q0 • helical-staircase	Reimagine this staircase as an ancient DNA double-helix sculpture, evoking genetic strands intertwining upward.	0.212	0	4	1	6	—
0101 • L5Q1 • extrusion-2020	Reinterpret the extrusion with a futuristic, sci-fi spacecraft-panel aesthetic.	0.273	3	0	0	8	—
0028 • L5Q3 • parametric-spiral-stairs	Make the spiral staircase feel like it belongs in a haunted, decaying gothic mansion.	0.333	3	2	0	6	—
0064 • L5Q1 • tool-shelf-for-bench	Give this tool shelf an organic, nature-inspired look, as if it grew from a tree trunk with wood-grain surfaces and root-like supports.	0.333	2	3	0	6	—
0110 • L5Q2 • cam-profile	Transform the cam profile into an organic, biomechanical form, as if it were grown rather than machined.	0.333	2	4	1	4	—
0028 • L5Q1 • parametric-spiral-stairs	Reimagine the spiral staircase as a magical DNA helix, evoking the structure of life itself.	0.364	0	6	1	4	—
0060 • L5Q2 • mini-storage-drawers	Give this storage unit a futuristic, high-tech sci-fi appearance.	0.364	4	0	0	7	—
0060 • L5Q3 • mini-storage-drawers	Make this cabinet feel playful and toy-like, as if designed for a child's room.	0.364	3	2	0	6	—
0101 • L5Q3 • extrusion-2020	Transform the extrusion into a luxury architectural element, as if used in high-end furniture design.	0.364	3	2	0	6	—
0095 • L5Q2 • helical-staircase	Give the staircase a futuristic sci-fi spaceship feel, as if it belongs aboard an interstellar vessel.	0.424	4	1	0	6	—
0095 • L5Q3 • helical-staircase	Reinterpret the staircase in an organic, nature-inspired style, as though it grew like a twisting vine or seashell spiral.	0.424	3	3	2	3	—
0110 • L5Q0 • cam-profile	Reimagine this cam profile as an industrial steampunk mechanism, like a component salvaged from a Victorian-era engine.	0.424	2	4	0	5	—
0011 • L5Q1 • curved-corner-bracket	Make the bracket feel organic and biomimetic, as if it grew naturally like a tree branch or bone joint.	0.432	3	3	0	5	—
0028 • L5Q0 • parametric-spiral-stairs	Transform this staircase into an elegant industrial-chic design, as if built from aged metal in a modern loft.	0.455	4	2	0	5	风格阈值存在 false-pass 风险
0060 • L5Q1 • mini-storage-drawers	Transform this cabinet into a cozy, warm wooden apothecary chest look.	0.455	5	0	1	5	—
0101 • L5Q2 • extrusion-2020	Give the extrusion an organic, bio-inspired appearance, as if grown rather than machined.	0.485	2	6	0	3	—
0005 • L5Q2 • ceiling-mounting-bracket	Transform the bracket to look organic, as if it grew naturally rather than being machined.	0.485	3	4	1	3	—
0060 • L5Q0 • mini-storage-drawers	Reimagine this storage cabinet with an industrial workshop aesthetic, like something from a machine shop toolbox.	0.500	5	1	1	4	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0129 • L5Q1 • neo-strip	Transform the strip's look to evoke natural wood and organic craftsmanship, like a handmade artisan light fixture.	0.515	5	1	0	5	—
0011 • L5Q0 • curved-corner-bracket	Reinterpret this bracket in an industrial gothic style, as if forged in a medieval blacksmith's shop.	0.545	6	0	0	5	—
0028 • L5Q2 • parametric-spiral-stairs	Give the staircase a royal, opulent palace feel fit for a grand ballroom entrance.	0.545	4	4	0	3	—
0061 • L5Q2 • customizable-furniture-minis	Reimagine the armchair with a futuristic, sci-fi aesthetic.	0.545	6	0	1	4	—
0064 • L5Q3 • tool-shelf-for-bench	Make this tool shelf feel futuristic and high-tech, like something out of a sci-fi workshop.	0.545	6	0	0	5	—
0110 • L5Q3 • cam-profile	Make the cam profile feel ancient and weathered, like an artifact excavated from an old ruin.	0.545	5	2	0	4	—
0129 • L5Q0 • neo-strip	Reimagine this LED strip channel with a sleek cyberpunk aesthetic, like something from a neon-lit futuristic cityscape.	0.545	6	0	0	5	—
0061 • L5Q1 • customizable-furniture-minis	Give the armchair a plush, cozy, and inviting look as if made from soft pillows.	0.576	6	1	0	4	—
0101 • L5Q0 • extrusion-2020	Give the extrusion a heavy-duty industrial look, like structural steel used in factory machinery.	0.576	6	1	0	4	—
0064 • L5Q0 • tool-shelf-for-bench	Reimagine this tool shelf with an industrial workshop aesthetic, as if it were welded from raw steel plate in a metal fabrication shop.	0.606	5	3	0	3	—
0110 • L5Q1 • cam-profile	Give the cam a futuristic sci-fi appearance, as if it were a precision component from an advanced spacecraft engine.	0.606	6	1	0	4	—
0011 • L5Q3 • curved-corner-bracket	Transform the bracket into a steampunk-inspired mechanical part, full of brass gears and rivets.	0.614	6	2	0	3	—
0005 • L5Q1 • ceiling-mounting-bracket	Give the bracket a sleek, minimalist modern design look, as if it were made for a high-end smart home device.	0.636	7	0	0	4	—
0061 • L5Q3 • customizable-furniture-minis	Turn the armchair into a rustic, cabin-style piece of furniture.	0.667	7	1	0	3	—
0061 • L5Q0 • customizable-furniture-minis	Transform this armchair into a sleek Scandinavian minimalist design.	0.705	7	1	0	3	—
0129 • L5Q3 • neo-strip	Redesign this component with an elegant, minimalist luxury feel, as if it belongs in a high-end modern home.	0.727	8	0	0	3	luxury/premium 判定疑似过宽
0005 • L5Q3 • ceiling-mounting-bracket	Make the bracket look armored and protective, like it's built to withstand harsh conditions.	0.758	8	1	0	2	—
0005 • L5Q0 • ceiling-mounting-bracket	Redesign the bracket with an industrial heavy-duty aesthetic, as if it belongs in a factory or warehouse.	0.788	8	1	0	2	—
0064 • L5Q2 • tool-shelf-for-bench	Transform this tool shelf into a sleek, minimalist Scandinavian-style storage piece.	0.788	8	1	0	2	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0129 • L5Q2 • neo-strip	Give this socket strip a stealthy, military-grade tactical look.	0.788	8	2	0	1	—
0011 • L5Q2 • curved-corner-bracket	Redesign the bracket to feel lightweight and aerodynamic, like a component from a race car or aircraft.	0.795	8	2	0	1	—

Level 6 任务矩阵

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0095 • L6Q1 • helical-staircase	Set the helical staircase in an outdoor garden courtyard: surround the base with a circular stone patio, add potted plants around the perimeter, a low wall or fence in the background, warm string lights wound around the central column, and a sun/sky lighting setup.	0.015	0	1	0	10	—
0095 • L6Q3 • helical-staircase	Create a fantasy tower library scene with the staircase as the centerpiece: add curved bookshelves lining the surrounding circular wall, a small round reading table with a chair on the ground floor, floating candle lights near the upper steps, and a stone floor beneath the staircase base.	0.015	0	1	6	4	—
0011 • L6Q2 • curved-corner-bracket	Build a furniture assembly scene: use the curved corner bracket to reinforce the corner joint between a wooden table leg and the underside of a tabletop, add a second identical bracket on an adjacent corner, place the table on a wooden floor with a rug beneath it, and add a small toolbox with a screwdriver nearby as if mid-assembly.	0.091	0	3	2	6	—
0110 • L6Q2 • cam-profile	Set the cam profile inside an engine valve-timing assembly scene: place it on a rotating camshaft supported by two engine block bearing mounts, add a valve spring and valve stem above contacting the cam lobe, include a timing belt or chain looping around a sprocket attached to the shaft, and surround the assembly with a simplified engine block housing.	0.152	0	4	1	6	—
0129 • L6Q3 • neo-strip	Place the strip channel in an outdoor patio decoration scene: mount it along the underside of a wooden pergola beam, add a patio table with two chairs beneath it, string lights draping between posts, potted plants at the base of the posts, and warm evening lighting overall.	0.156	0	2	2	7	—
0011 • L6Q3 • curved-corner-bracket	Design an outdoor pergola support scene: mount the curved corner bracket at the junction of a vertical wooden post and a horizontal roof beam outdoors, add three more brackets at other post-beam junctions, surround the pergola with potted plants, lay a stone patio floor beneath, and light the scene with natural daytime sunlight and soft shadows.	0.169	1	1	0	9	—
0110 • L6Q1 • cam-profile	Display the cam profile as an educational exhibit on a museum pedestal: place it upright on a rotating display stand, add a glass case enclosure around it, a descriptive placard in front, museum spotlight lighting from above, and a polished floor beneath the pedestal.	0.169	1	2	0	8	—
0005 • L6Q2 • ceiling-mounting-bracket	Build an installation-in-progress scene: show the bracket mounted to a wooden ceiling joist with two visible screws driven into the round bosses, a cordless drill floating nearby as if just used, a small pile of wood shavings and screws on the floor below, and a work light illuminating the joist from the side.	0.182	0	4	1	6	—
0028 • L6Q1 • parametric-spiral-stairs	Set the spiral staircase into an outdoor garden courtyard scene: place it on a circular stone platform, surround it with potted plants at the base, add a ground plane with grass texture/color, a soft sky-colored background, and two small lantern lights flanking the entrance at the bottom of the stairs.	0.182	0	4	1	6	—
0028 • L6Q2 • parametric-spiral-stairs	Build a sci-fi hangar scene around the spiral staircase: add a metallic floor grating beneath it, a large cylindrical elevator shaft wall around the stairs with glowing blue trim lights, an overhead ceiling with hanging industrial lamps, and a robotic maintenance drone floating near the middle of the staircase.	0.195	0	7	0	4	—
0101 • L6Q0 • extrusion-2020	Build a modular workbench frame scene using multiple copies of this 2020 extrusion: arrange four uprights as legs and four horizontal pieces as a rectangular top frame and a lower shelf frame, connected with simple corner bracket cubes at each joint. Place the assembled frame on a concrete floor, add a plywood panel resting on top of the upper frame to act as a tabletop, and light the scene with a soft overhead studio light.	0.205	1	3	2	5	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0005 • L6Q1 • ceiling-mounting-bracket	Create a hardware store display scene for the bracket: place it standing on a wooden shelf, add two more identical brackets lined up beside it, a small price tag card in front, a cardboard box behind stacked with similar brackets, and overhead spotlight illumination highlighting the display.	0.212	1	2	2	6	—
0064 • L6Q2 • tool-shelf-for-bench	Build a small outdoor shed scene around the tool shelf: attach it to a wooden shed wall, place a potted plant on the ground to one side, add a garden hose coiled on a hook near the shelf, put a few small flower pots on the front-most shelf level, and light the scene with soft daylight from a window.	0.221	1	3	0	7	—
0101 • L6Q3 • extrusion-2020	Compose a robotics kit unboxing scene: lay the extrusion piece on a foam-lined cardboard box tray, surround it with other kit parts such as wheels, a small circuit board, screws, and a coiled cable, place an instruction manual booklet next to the box, and light the scene with soft window daylight coming from one side.	0.234	0	4	1	6	—
0005 • L6Q3 • ceiling-mounting-bracket	Compose a warehouse ceiling installation scene: suspend the bracket from a metal ceiling grid using a threaded rod through its boss hole, attach a length of pipe or conduit resting in its C-channel running off to the side, add a second bracket further down the same pipe run, place a rolling ladder below for context, and light the scene with industrial overhead fluorescent fixtures.	0.242	0	6	0	5	—
0061 • L6Q1 • customizable-furniture-minis	Set up the armchair as part of a miniature furniture showroom display: place it on a raised platform/pedestal, add a small price tag or label card floating near it, position two other miniature chair variants nearby for comparison, and add soft studio-style spotlights above the display.	0.242	0	4	1	6	—
0028 • L6Q3 • parametric-spiral-stairs	Create a museum atrium display scene featuring the spiral staircase: place it in the center of a large open room with a glossy tiled floor, add glass display cases with small artifacts along the walls, a skylight ceiling letting in bright daylight, a museum information placard on a stand near the staircase base, and a couple of small human figure silhouettes viewing the exhibit.	0.247	1	5	2	3	—
0101 • L6Q1 • extrusion-2020	Create a hardware store display scene: stand the extrusion piece upright in a pegboard-backed display bin alongside a small pile of T-nuts, bolts, and corner brackets scattered at its base, add a printed price tag/label card leaning against the bin, place a shelf beneath everything, and light the scene with bright retail ceiling lights.	0.247	1	3	1	6	—
0011 • L6Q1 • curved-corner-bracket	Create a hardware store display scene: place five identical curved corner brackets neatly arranged in a row on a metal display shelf, add a price tag label hanging in front, a cardboard box of extra brackets beside the shelf, and warm overhead spotlight lighting illuminating the display.	0.260	0	5	1	5	—
0060 • L6Q3 • mini-storage-drawers	Build a garage storage scene: mount the mini storage drawer cabinet on a wall-mounted shelf, place a toolbox on the floor below it, hang a wrench and pliers on a nearby pegboard, add a rolling stool beside the shelf, and light the space with a single hanging garage bulb.	0.260	2	1	2	6	—
0060 • L6Q2 • mini-storage-drawers	Create a small retail display scene: place the mini storage drawer cabinet on a shop counter, add a price tag hanging from it, a cash register beside it on the counter, a shopping bag on the floor nearby, and shelf lighting overhead illuminating the display.	0.273	0	5	0	6	—
0064 • L6Q1 • tool-shelf-for-bench	Place the tool shelf into a garage scene: set it on a shelf bracket mounted to a brick wall, add a concrete floor beneath, park a toolbox on the floor below the shelf, lean a broom against the wall nearby, and add a ceiling light fixture above casting light down onto the shelf.	0.273	0	6	0	5	—
0064 • L6Q3 • tool-shelf-for-bench	Design a kitchen-adjacent utility nook scene featuring the tool shelf repurposed for storage: mount it above a small side table, add a stack of folded towels on the middle level, a few mason jars with lids on the rear level, a woven basket on the floor beneath the table, and a warm wall sconce light above the shelf for ambiance.	0.273	0	5	0	6	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0061 • L6Q3 • customizable-furniture-minis	Build a miniature outdoor patio scene featuring the armchair: place it on a small wooden deck platform, add a miniature side table with a drink on it next to the armchair, a string of tiny fairy lights strung overhead, a small potted flower plant nearby, and a sun/sky-toned background to suggest an outdoor setting.	0.286	1	3	0	7	—
0129 • L6Q2 • neo-strip	Create a computer case interior scene featuring this LED strip channel: attach it along the inside edge of a PC case side panel, add a motherboard with visible components, two case fans, cable routing channels, and a tempered glass side panel showing the illuminated interior.	0.286	1	3	0	7	—
0060 • L6Q1 • mini-storage-drawers	Set the mini storage drawer cabinet on a workbench in a craft room. Fill some of its compartments with visible small items like beads or screws, add a pair of scissors and a spool of thread on the bench next to it, hang a pegboard with tools on the wall behind, and light the scene with an overhead work lamp.	0.295	1	6	0	4	—
0028 • L6Q0 • parametric-spiral-stairs	Place the spiral staircase inside a lighthouse tower interior: add a cylindrical outer wall enclosing the stairs with small round windows, a floor at the base, a landing platform with a railing at the top, and a warm lantern light source at the very top of the tower shining downward through the stairwell.	0.299	1	5	1	4	封闭圆筒遮挡内部; 存在 false pass
0129 • L6Q1 • neo-strip	Integrate the strip channel into a home theater ambient lighting scene: mount it along the base of a TV stand, place a television above it, add a media console below with a game console and remote, dim ambient room lighting, and a sofa positioned facing the TV in the background.	0.299	2	3	2	4	—
0110 • L6Q3 • cam-profile	Create a workbench repair scene around the cam profile: place it lying on a wooden workbench next to a set of hand tools (wrench and screwdriver), add a small oil can and a rag beside it, an overhead work lamp shining down, and a mechanic's stool positioned in front of the bench.	0.325	2	2	2	5	—
0061 • L6Q2 • customizable-furniture-minis	Create a cozy miniature reading nook scene: place the armchair near a small round window with a sill, add a tiny bookshelf filled with miniature books to one side, a small footstool in front of the armchair, a warm floor lamp beside it, and a cat figurine curled up on the seat cushion.	0.338	1	4	1	5	—
0095 • L6Q2 • helical-staircase	Build a lighthouse interior scene around the staircase: enclose it inside a cylindrical stone tower wall, add small round window openings at intervals in the wall aligned with the treads, a wooden entry door at the base, a lantern room structure with a light beacon at the top, and dim warm interior lighting.	0.364	0	7	0	4	—
0101 • L6Q2 • extrusion-2020	Set up a 3D printer frame corner scene: use the extrusion as one vertical frame rail, attach a corner bracket plate at its base connecting it to two other extrusion rails forming an L-shaped base, mount a small stepper motor model at the bottom of the vertical rail, add a drive belt loop running along one side, and place the whole assembly on a workshop table with visible wood grain.	0.364	1	5	0	5	—
0061 • L6Q0 • customizable-furniture-minis	Compose a dollhouse living room scene around the miniature armchair: place it on a small patterned rug, add a tiny wooden side table with a miniature lamp beside the armchair, put a small bookshelf against the back wall, and include a miniature potted plant in the corner. Light the scene with a warm ceiling light fixture.	0.386	1	5	1	4	—
0005 • L6Q0 • ceiling-mounting-bracket	Set up the ceiling mounting bracket in a finished room ceiling scene: attach it flush to a flat ceiling plane with a screw through its round boss, hang a cylindrical rod or conduit segment clipped into its C-shaped channel, add a ceiling-mounted light fixture nearby for context, and include a soft ambient light so the room feels naturally lit.	0.409	1	5	2	3	—
0011 • L6Q0 • curved-corner-bracket	Set up a workshop wall-mounting scene: attach the curved corner bracket to a vertical wooden wall panel and a horizontal wooden shelf so it forms an L-shaped support joint, add several wood screws inserted through its holes, place a pegboard with hanging hand tools behind the wall panel, and add a workbench below with scattered metal shavings.	0.429	2	4	2	3	—

Case	Query	Mean	Full	Partial	Zero	RF	审计风险
0110 • L6Q0 • cam-profile	Mount the cam profile onto a functioning cam-and-follower mechanism: add a horizontal steel shaft through its central hole, place a small roller/follower resting on the cam's edge with a vertical push-rod above it, set the assembly on a metal machine base plate, and add two pillow-block bearings supporting the shaft on either side of the cam.	0.468	2	5	0	4	—
0095 • L6Q0 • helical-staircase	Place the helical staircase inside a modern multi-story atrium: add a floor slab beneath it, a ceiling skylight above with a circular opening for the stairs, glass railing panels following the outer edge of the treads, and a pair of pendant lights hanging near the top landing.	0.470	3	5	0	3	—
0060 • L6Q0 • mini-storage-drawers	Place the mini storage drawer cabinet on top of a wooden desk in a home office. Add a desk lamp beside it casting warm light, a cup of pens next to the cabinet, a laptop further back on the desk, and a chair pushed in beneath the desk.	0.481	4	3	0	4	—
0064 • L6Q0 • tool-shelf-for-bench	Mount the tool shelf onto the back wall above a wooden workbench: add a rectangular workbench surface positioned directly below the shelf, a plain wall panel behind it, and hang a row of hand tools (wrench, screwdriver, hammer) on the shelf's pegboard-style holes so they appear to dangle through the perforations. Add a single overhead work light illuminating the bench area.	0.532	4	4	1	2	—
0129 • L6Q0 • neo-strip	Set up a workbench scene around this NeoPixel strip channel: place it on a wooden workbench surface, add a spool of addressable LED strip lights nearby, a soldering iron resting on a stand, a small pair of wire cutters, and a desk lamp positioned above to illuminate the work area.	0.610	5	3	0	3	对象存在不等于关系/尺度成立

附录 B · Agent fan-out 与证据清单

底稿	角色	大小
sol_reviews/level_1.md	GPT-5.6 Sol · Level 1 独立审计	9.9 KiB
sol_reviews/level_2_retry.md	GPT-5.6 Sol · Level 2 独立审计	11.3 KiB
sol_reviews/level_3.md	GPT-5.6 Sol · Level 3 独立审计	10.6 KiB
sol_reviews/level_4.md	GPT-5.6 Sol · Level 4 独立审计	13.0 KiB
sol_reviews/level_5.md	GPT-5.6 Sol · Level 5 独立审计	14.9 KiB
sol_reviews/level_6.md	GPT-5.6 Sol · Level 6 独立审计	14.1 KiB
agent_levels_1_2.md	跨层全量交叉审计 · L1/L2	22.8 KiB
agent_levels_3_4.md	跨层全量交叉审计 · L3/L4	26.0 KiB
agent_levels_5_6.md	跨层全量交叉审计 · L5/L6	17.0 KiB
sol_reviews/verifier_false_negative.md	Verifier false-negative 专项	6.2 KiB
sol_reviews/verifier_false_positive.md	Verifier false-positive 专项	10.0 KiB

关键可复核产物

- verifier_recheck_sonnet5.json：5 个 case × 5 次 Sonnet 5 重复验证的 raw response 与 score。
- sol_reviews/diagnostic_l1_sofa.png：sofa 高度/长度前后对照。
- sol_reviews/diagnostic_l1_contact.png：L1 可见性诊断。
- report/cases.html：2,640 条 evaluation 的交互筛选，含 prediction/script/evaluation/error 与六视图路径。
- report/report_manifest.json：报告范围、页面和生成时间。

最终交付说明

本 PDF 是可移植的完整叙事版，包含核心统计、每层/每模型代表、失败分类、verifier 证据、60-task 矩阵与 agent provenance。2,640 条逐案明细保留在 ZIP 的 HTML snapshot；原始六视图和脚本数据体量为数 GiB，没有复制进 ZIP。